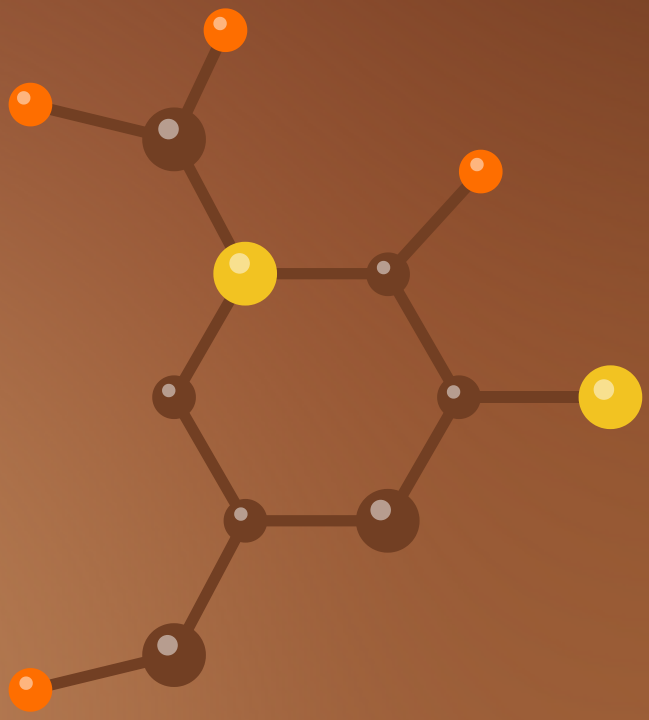
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success



ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Halogen Derivatives

English Medium

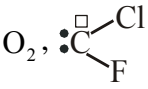
HALOGEN DERIVATIVES

EXERCISE # O-1 (MAINS ORIENTED)

1. Identify set of electrophiles :

(A) CO_2, $\overset{\oplus}{C}H_3$, $:\overset{\square}{C}H_2$, Br_2

(B) HOH, SO_3, $:\overset{\square}{C}Cl_2$, $Cl^{\oplus}$

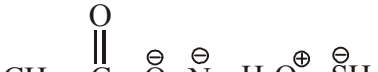
(C) SO_2, $CH_3-\ddot{O}H$, $\overset{\oplus}{N}O_2$, 

(D) $H-\overset{\oplus}{C}=O$, $\overset{\oplus}{N}=O$, $Ph-\overset{\oplus}{C}H_2$, $\ddot{N}H_3$

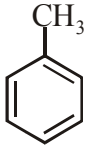
HD0001

2. Identify set of nucleophiles :

(A) $\overset{\ominus}{Cl}$, $\overset{\ominus}{O}H$, $\overset{\ominus}{R}$, $:\overset{\square}{C}H_2$

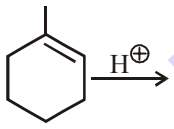
(B) 

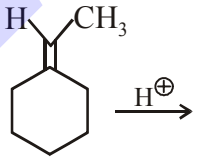
(C) $CH_2=CH_2$, $CH_3-\ddot{N}H_2$, $CH_3-CH_2-\ddot{O}H$, 

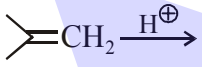
(D) $\ddot{H}^{\ominus}$, $:\overset{\ominus}{C}N$, CS_2, 

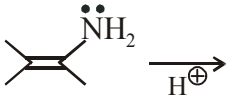
HD0002

3. Which of the following will form 2° carbocation?

(A) 

(B) 

(C) 

(D) 

HD0003

4. Incorrect statement about carbocation is :

(A) It is lewis acid

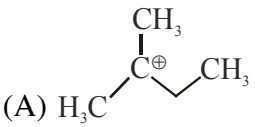
(B) It has 6 electrons in valency shell

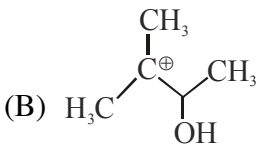
(C) It is electrophile

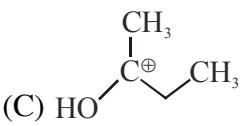
(D) It is always trigonal planer

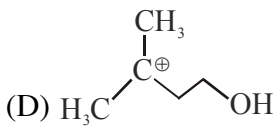
HD0004

5. Which of the following carbocation is most stable ?

(A) 

(B) 

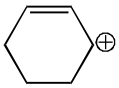
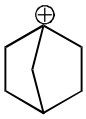
(C) 

(D) 

HD0005

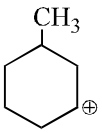
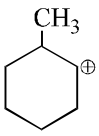
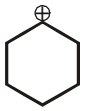
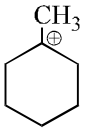
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\O1_Ex#O-1.p65"

6. Which carbocation is least likely to be formed as an intermediate ?

(A) $(C_6H_5)_3\overset{\oplus}{C}$ (B)  (C)  (D) $CH_3-\overset{\oplus}{C}H_2$

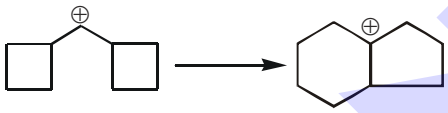
HD0006

7. Which one of the following carbocation would you expect to rearrange :

(A)  (B)  (C)  (D) 

HD0007

8. How many 1,2-shifts are involved during the course of following reaction :

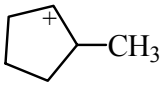
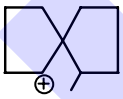
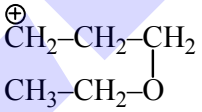
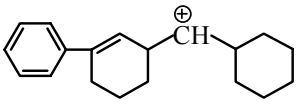


(A) 1 (B) 2 (C) 3 (D) 4

HD0008

9. How many following carbocation undergo re-arrangement -

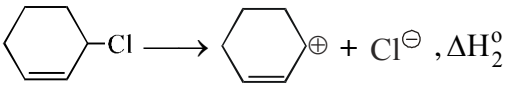
(a) $CH_3CH_2CH_2^+$ (b) $(CH_3)_2\overset{+}{C}HCHCH_3$ (c) $(CH_3)_3C\overset{+}{C}HCH_3$ (d) $(CH_3CH_2)_3CCH_2^+$

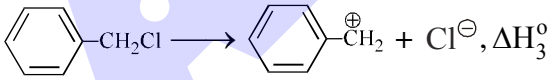
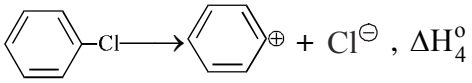
(e)  (f)  (g) $\overset{\oplus}{C}H_2-CH_2-CH_2$  (h) 

(A) 5 (B) 8 (C) 6 (D) 7

HD0009

10. For the reactions

(I)  (II) 

(III)  (IV) 

The correct decreasing order of enthalpies of reaction for producing carbocation is :

(A) $\Delta H_1^o > \Delta H_2^o > \Delta H_3^o > \Delta H_4^o$

(B) $\Delta H_4^o > \Delta H_1^o > \Delta H_2^o > \Delta H_3^o$

(C) $\Delta H_3^o > \Delta H_2^o > \Delta H_1^o > \Delta H_4^o$

(D) $\Delta H_2^o > \Delta H_1^o > \Delta H_4^o > \Delta H_3^o$

HD0010

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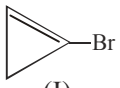
11.  , which is not the correct statement :

(I)

(A) I is more soluble in water than bromocyclopropane

(B) I gives pale yellow ppt. on addition with aq. $AgNO_3$

(C) I is having lower dipole moment than bromocyclopropane

(D) I is more ionic than 

(I)

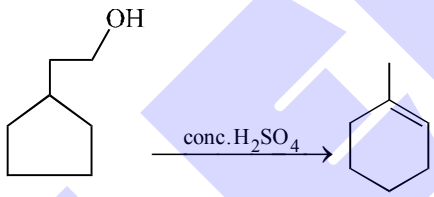
HD0011

12. A solution of (–) –1–chloro–1–phenylethane in toluene racemises slowly in the presence of a small amount of $SbCl_5$, due to the formation of :-

(A) carbanion (B) Carbene (C) carbocation (D) free radical

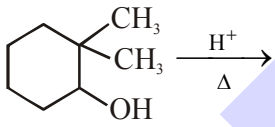
HD0012

13. How many 1,2-Shifts of carbocation intermediate are involved during the course of following reaction :



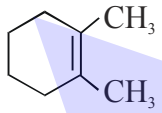
(A) 1 (B) 2 (C) 3 (D) 4

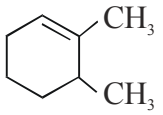
HD0013

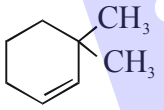
14. 

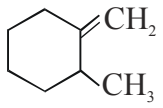
(X) (Major product)

Major product (X) is :

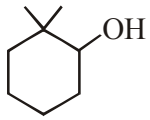
(A) 

(B) 

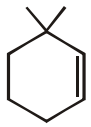
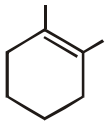
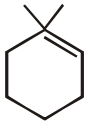
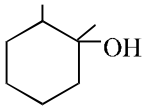
(C) 

(D) 

HD0014

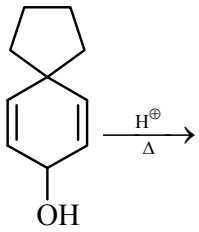
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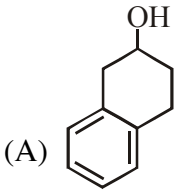
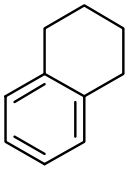
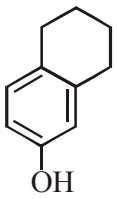
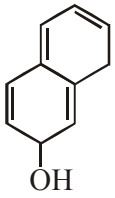
P. The product P is :

(A)  (B)  (C)  (D) 

HD0015

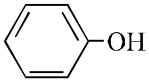
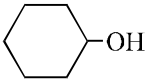
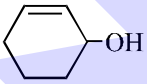
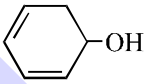
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65"

16.  Product ; Product is :

(A)  (B)  (C)  (D) 

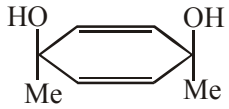
HD0016

17. Among the given compounds, the correct order of rate of dehydration is :

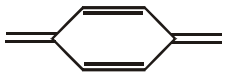
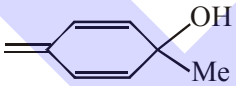
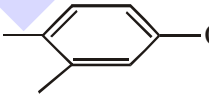
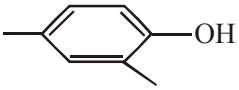
(I)  (II)  (III)  (IV) 

(A) I < II < III < IV (B) II < III < IV < I (C) I < III < IV < II (D) I < II < III = IV

HD0017

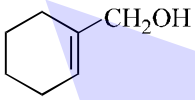
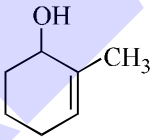
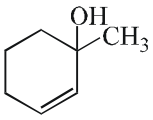
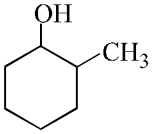
18.  Major product -

Major product is :

(A)  (B)  (C)  (D) 

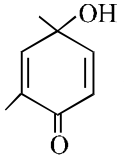
HD0018

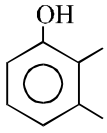
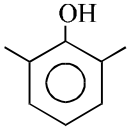
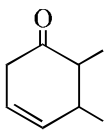
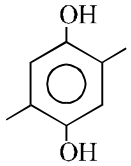
19. Identify the correct order of rate of dehydration when given compounds are treated with conc. H_2SO_4:

(P)  (Q)  (R)  (S) 

(A) P > Q > R > S (B) Q > P > R > S (C) R > Q > P > S (D) R > Q > S > P

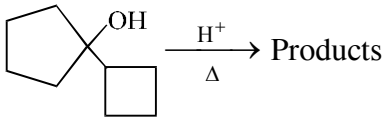
HD0019

20.  X ; X is :

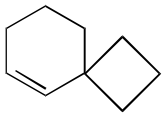
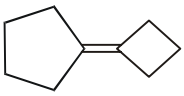
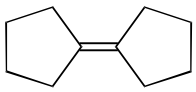
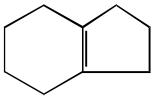
(A)  (B)  (C)  (D) 

HD0020

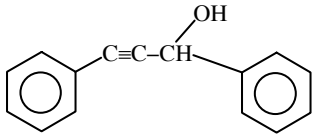
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Wbd.#Hdo-Al.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtrO-I.p65"

21. 

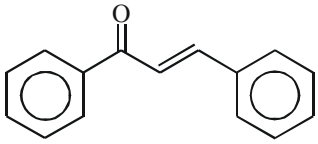
Major products is :

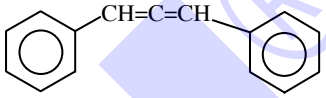
(A) 
(B) 
(C) 
(D) 

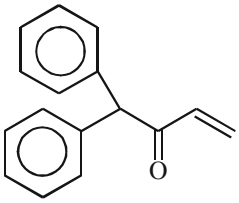
HD0021

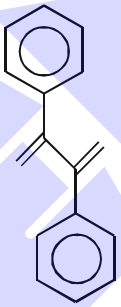
22. 

$\xrightarrow{H^+}$ Product is :

(A) 

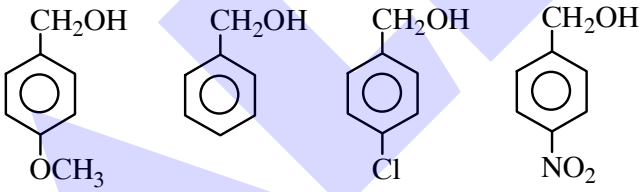
(B) 

(C) 

(D) 

HD0022

23. What is the decreasing order of rate of reaction with HBr for the following benzyl alcohol and its derivative :



(A) (B) (C) (D)

(A) A > C > D > B (B) A > B > D > C (C) D > C > B > A (D) A > B > C > D

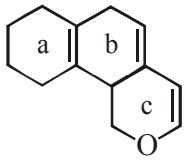
HD0023

24. Which will dehydrate at fastest rate by H_3PO_4 :

(A) 2-methyl butan-2-ol (B) 3-methyl butan-2-ol

(C) Butan-1-ol (D) 2-methyl butan-1-ol

HD0024

25. 

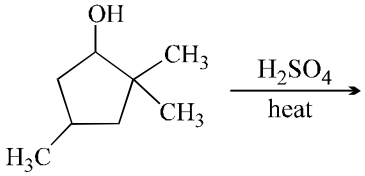
The double bond which is most reactive towards attack of electrophile :

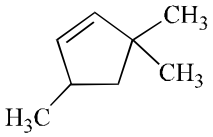
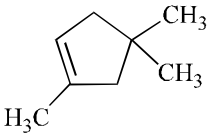
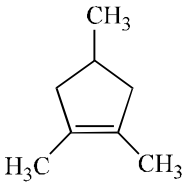
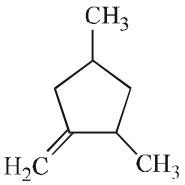
(A) a (B) b (C) c (D) None

HD0025

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.& Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtMO-1.p65"

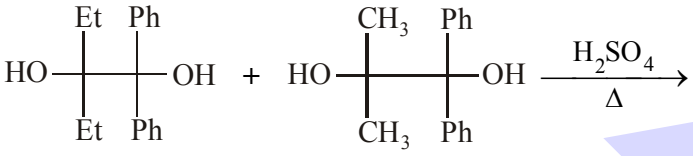
26. The major product formed in the following reaction is :



(A)  (B)  (C)  (D) 

HD0026

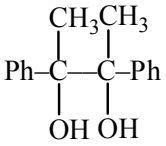
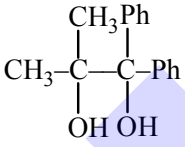
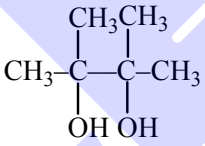
27. How many products are obtained in the given reaction :



(A) 1 (B) 2 (C) 3 (D) 4

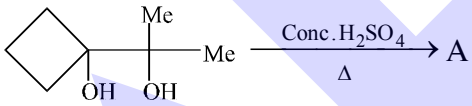
HD0027

28. Compare rate of reaction towards pinacol pinacolone rearrangement.

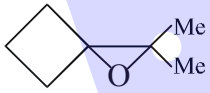
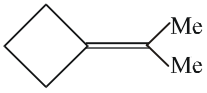
 (I)  (II)  (III)

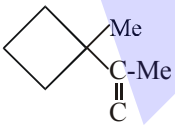
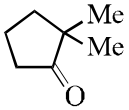
(A) II > III > I (B) III > II > I (C) II > I > III (D) I > II > III

HD0028

29. 

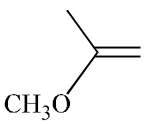
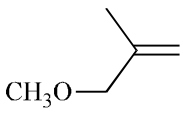
Product A is :

(A)  (B) 

(C)  (D) 

HD0029

30. What is the order of reactivity with HBr :

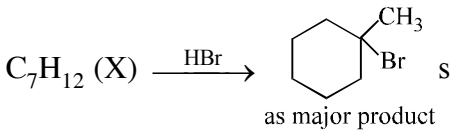
(x)  (y)  (z) 

(A) x > y > z (B) y > x > z (C) z > y > x (D) y > z > x

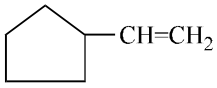
HD0030

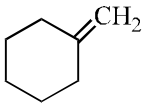
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hydr.-Carb.-Oxi.-Red.-Alcl & Er.-Hydr.-Carb.-Oxi.-Red.-Alcl & Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtraO-1.p65

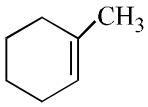
31. In the given reaction

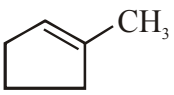


(X) can not be :

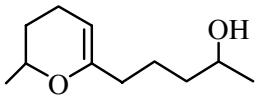
(A) 

(B) 

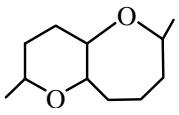
(C) 

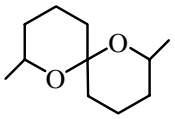
(D) 

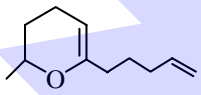
HD0031

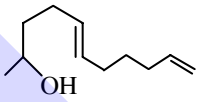
32. 

'P' (Major Product)

(A) 

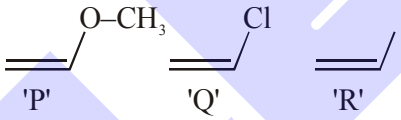
(B) 

(C) 

(D) 

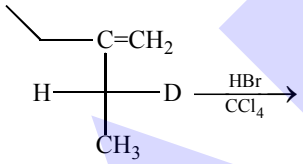
HD0032

33. Arrange the following compounds in decreasing order of electrophilic addition :



(A) P > Q > R (B) P > R > Q (C) R > P > Q (D) R > Q > P

HD0033

34. 

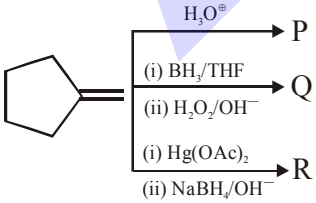
Product (without rearrangement of carbocation)

What is stereochemistry of product :

(A) Racemic mixture (B) Optically inactive

(C) Mixture of diastereomers (D) Meso product

HD0034

35. 

Correct statement regarding products P, Q & R

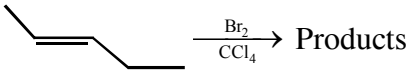
(A) Product P & R are same (B) Product Q & R are same

(C) P & Q are functional isomers (D) Product P, Q & R all are different

HD0035

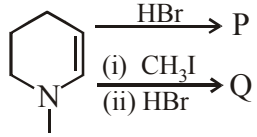
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word#Hdc-Al-& Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65"

36. Select the incorrect statement about the product mixture in the following reaction :

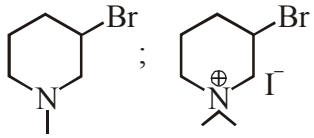


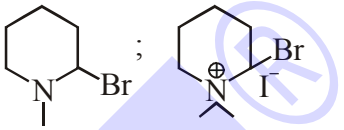
(A) It is optically active
(B) It is racemic mixture
(C) It is a resolvable mixture
(D) It is a mixture of erythro compounds

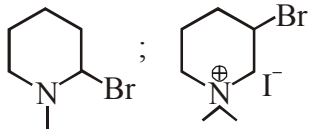
HD0036

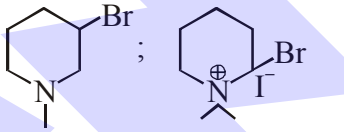
37. 

If P & Q are the major products then P & Q are respectively :

(A) 

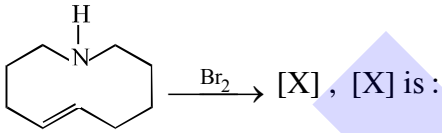
(B) 

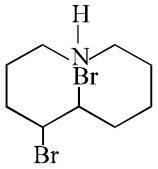
(C) 

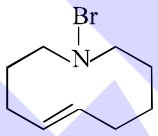
(D) 

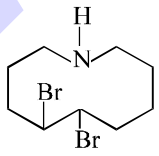
HD0037

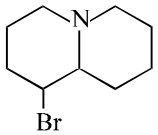
38. In the given reaction :



(A) 

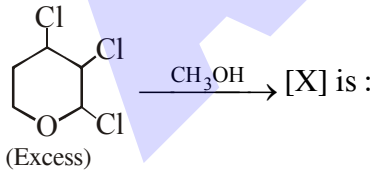
(B) 

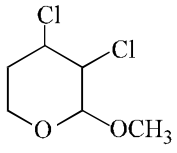
(C) 

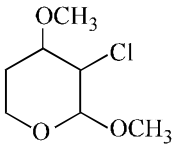
(D) 

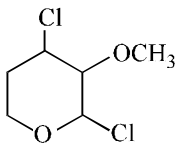
HD0038

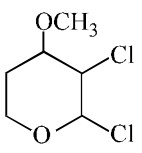
39. In the given reaction:



(A) 

(B) 

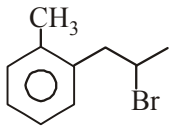
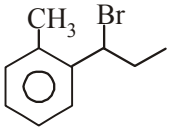
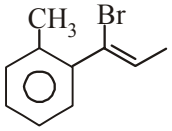
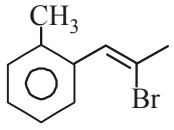
(C) 

(D) 

HD0039

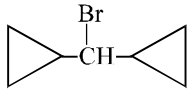
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & Et - Hydr - Carb.-Oxi. Red. Ald & Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65

40. Which compound undergoes hydrolysis by the S_N1 mechanism at the fastest rate?

(A)  (B)  (C)  (D) 

HD0040

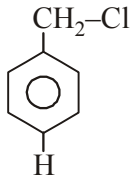
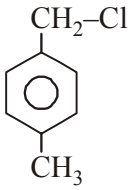
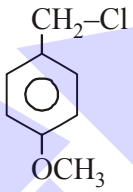
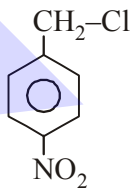
41. Arrange the following compounds in decreasing order of their reactivity for hydrolysis reaction

(I) $C_6H_5–CH_2–Br$ (II) $C_6H_5-\underset{\displaystyle Br}{\underset{|}{CH}}-C_2H_5$ (III)  (IV) 

(A) I > II > III > IV (B) IV > II > I > III (C) III > IV > II > I (D) IV > III > II > I

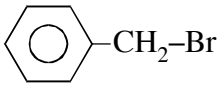
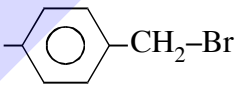
HD0041

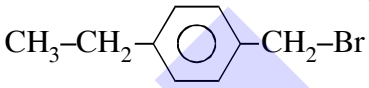
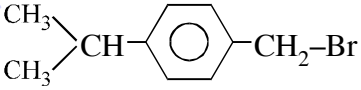
42. Which of the following is most reactive toward S_N1 reaction.

(A)  (B)  (C)  (D) 

HD0042

43. Arrange the following compounds in order of decreasing rate of hydrolysis for S_N1 reaction:

(I)  (II) 

(III)  (IV) 

(A) II > III > IV > I (B) IV > III > II > I (C) III > IV > II > I (D) I > II > III > I

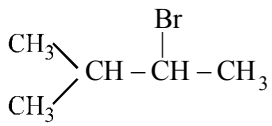
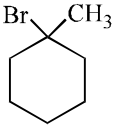
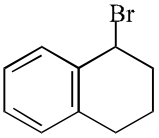
HD0043

44. Which one of the following compounds will give enantiomeric pair on treatment with HOH?

(A) $C_6H_5-\underset{\displaystyle C_2H_5}{\overset{\displaystyle C_2H_5}{\overset{|}{\underset{|}{C}}}}-I$ (B) $CH_3-\underset{\displaystyle C_2H_5}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-Br$ (C) $C_6H_5-\underset{\displaystyle D}{\overset{\displaystyle H}{\overset{|}{\underset{|}{C}}}}-Br$ (D) $C_2H_5-\underset{\displaystyle CH_3}{\overset{\displaystyle C_2H_5}{\overset{|}{\underset{|}{C}}}}-Br$

HD0044

45. Consider the S_N1 solvolysis of the following halides in aqueous formic acid:

(I)  (II)  (III) $C_6H_5-\underset{\displaystyle Br}{\underset{|}{CH}}-C_6H_5$ (IV) 

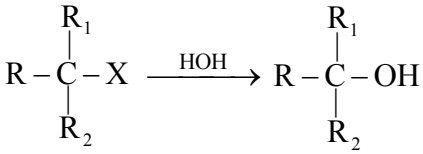
Decide decreasing order of reactivity of above alkyl halide?

(A) III > IV > II > I (B) II > IV > I > III (C) I > II > III > IV (D) III > I > II > IV

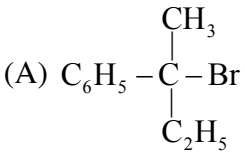
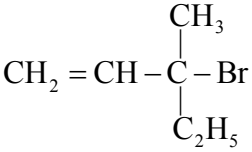
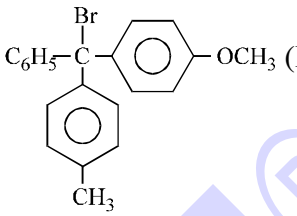
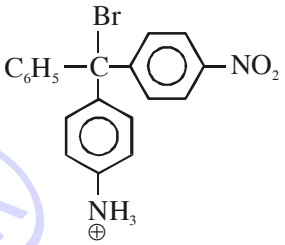
HD0045

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Al-& Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65"

46. For the given reaction

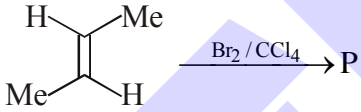


Which substrate will give maximum racemisation?

(A) 
(B) 
(C) 
(D) 

HD0046

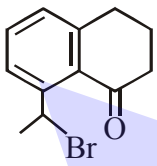
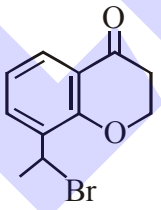
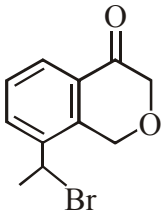
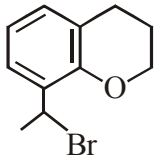
47. Select incorrect statements about the product (P) of the reaction :



(A) P is optically inactive due to internal compensation
(B) P is optically inactive due to the presence of plane of symmetry in the molecule
(C) The structure of P can have three optical isomers possible.
(D) P can have four possible optical isomers.

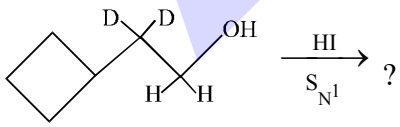
HD0047

48. Consider the following molecules :

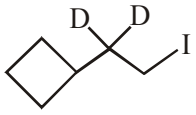
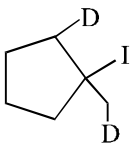
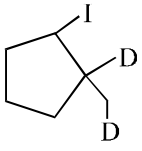
(I) 
(II) 
(III) 
(IV) 

The correct decreasing ease of hydrolysis of alkyl halide is :
(A) II > III > IV > I (B) II > IV > III > I (C) II > I > III > IV (D) IV > II > III > I

HD0048

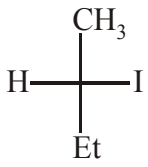
49. 

Major product is:

(A) 
(B) 
(C) 
(D) None of these

HD0049

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hydr.-Carb.-Oxi.-Red.-Alkl.& Ele.-Hydr.-Carb.-Oxi.-Red.-Alkl.& Ran-(OC)\Eng\01_Halogen Derivatives\01_ExrO-I.p65"

50.  $\xrightarrow{HOH}$ Products. (If 96% racemisation takes place)

Find out the correct statement about the reaction.

(A) Among the products 48% S and 48% R configuration containing molecules are present

(B) Among the products 50% S and 50% R configuration containing molecules are present

(C) Among the products 48% S and 52% R configuration containing molecules are present

(D) Among the products 52% S and 48% R configuration containing molecules are present

HD0050

51. In the given reaction the product [P] can be :

$CH_3–CH=CH–CH_2–OH \xrightarrow[S_N1']{HBr} [P]$

(A) $CH_3–CH=CH–CH_2–Br$

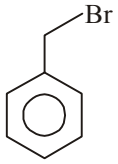
(B) $CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH=CH_2$

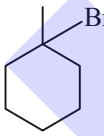
(C) $CH_2=CH–CH=CH_2$

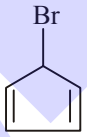
(D) $CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-CH_2-OH$

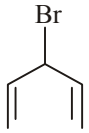
HD0051

52. Which of the following can not give S_N1 reaction easily?

(A) 

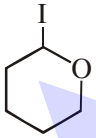
(B) 

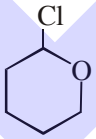
(C) 

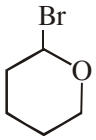
(D) 

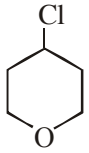
HD0052

53. Which one of the following compounds will be most reactive for S_N1 reactions ?

(A) 

(B) 

(C) 

(D) 

HD0053

54. Which of the following compounds is most rapidly hydrolysed by S_N1 mechanism ?

(A) C_6H_5Cl

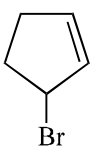
(B) $Cl–CH_2–CH=CH_2$

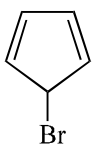
(C) $(C_6H_5)_3CCl$

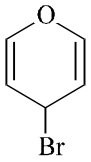
(D) $C_6H_5CH_2Cl$

HD0054

55. Among the bromides I–III given below, the order of reactivity in S_N1 reaction is:

(I) 

(II) 

(III) 

(A) III > I > II

(B) III > II > I

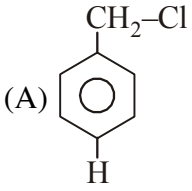
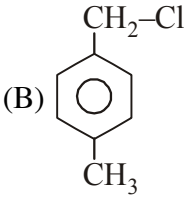
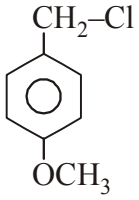
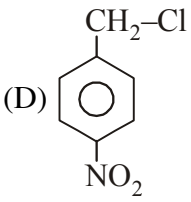
(C) II > III > I

(D) II > I > III

HD0055

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word Files\Halo-Al-& Et-Hydr.-Carb.-Oxi.-Red.-All.& Ran-(OC)\Eng\01_Halogen Derivatives\01_ExtO-1.p65"

56. Which of the following is most reactive toward S_N2.

(A)  (B)  (C)  (D) 

HD0056

57. For reaction $CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$

the rate of reaction is given by the expression :

(A) Rate = k $[CH_3Br]$ (B) Rate = k $[OH^-]$

(C) Rate = k $[CH_3Br][OH^-]$ (D) Rate = k $[CH_3Br]^{\circ}$ $[OH^-]^{\circ}$ **HD0057**

58. Select suitable reason for non-occurence of the following reaction.

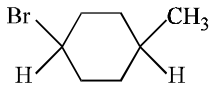
$Br^- + CH_3OH \longrightarrow BrCH_3 + OH^-$

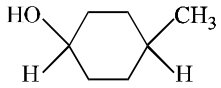
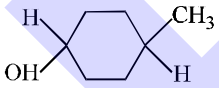
(A) Attacking nucleophile is stronger one

(B) Leaving group is a stronger base than nucleophile

(C) Alcohols are not good substrate for S_N reaction

(D) Hydroxide ions are weak bases **HD0058**

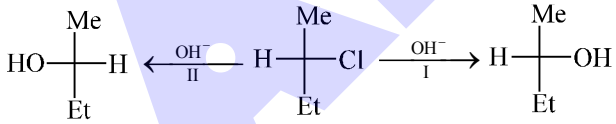
59.  + OH^- $\xrightarrow{S_N2}$ A ; A is :

(A)  (B)  (C) Both (D) None **HD0059**

60. The reactivity of 2-bromo-2-methylbutane (I), 1-bromopentane (II) and 2-bromopentane (III) towards S_N2 displacement is such that :

(A) I > II > III (B) I > III > II (C) II > III > I (D) II > I > III

HD0060

61. 

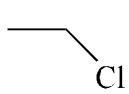
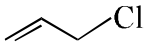
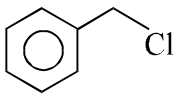
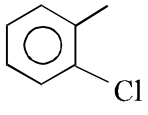
B A

Mechanism involved :

(A) I can't be S_N1 (B) II can't be S_N2 (C) I S_N1 & II S_N2 (D) I S_N2 & II S_N1

HD0061

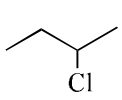
62. In which of the following replacement of Cl^- is most difficult ?

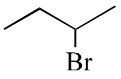
(A)  (B)  (C)  (D) 

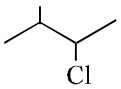
HD0062

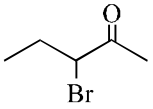
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc.& Et.-Hydr.-Carb.-Oxi.-Red.-Alld.& Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65

63. Arrange these compounds in order of increasing S_N2 reaction rate :


(I)


(II)

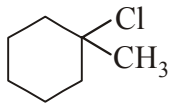
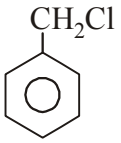
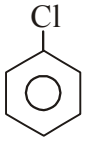
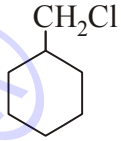

(III)


(IV)

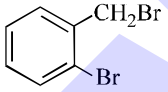
(A) III < I < II < IV (B) III < II < I < IV (C) IV < III < I < II (D) III < IV < I < II

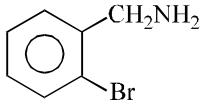
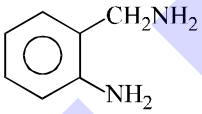
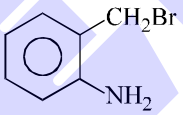
HD0063

64. Which reaction proceeds faster with NaI in DMSO ?

(A)  (B)  (C)  (D) 

HD0064

65. The major product in the given reaction:  $+ NH_3 \longrightarrow$

(A)  (B)  (C)  (D) All of these

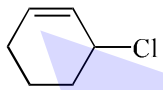
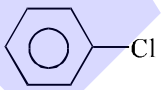
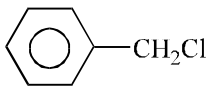
HD0065

66. The given compound $CH_3–O–CH_2–Br$ gives which one of the following reactions:

(A) Only S_N1 (B) Only S_N2

(C) S_N1 as well as S_N2 (D) E_1 only **HD0066**

67. Which will give white ppt. with $AgNO_3$?

(A)  (B)  (C)  (D) Both A & C

HD0067

68. Consider the following groups :

(I) –OAc (II) –OMe (III) $–OSO_2Me$ (IV) $–OSO_2CF_3$

The order of leaving group nature is:

(A) I > II > III > IV (B) IV > III > I > II (C) III > II > I > IV (D) II > III > IV > I

HD0068

69. When ethyl bromide is treated with moist Ag_2O, the main product is:

(A) Ethyl ether (B) Ethanol (C) Ethoxy ethane (D) All of these

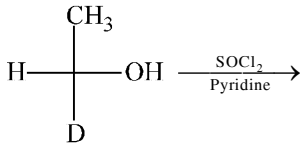
HD0069

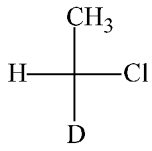
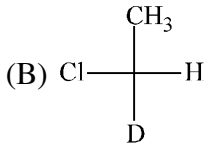
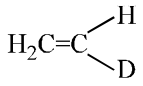
70. When ethyl bromide is treated with dry Ag_2O, the main product is:

(A) Ethyl ether (B) Ethanol (C) Ethoxy ethane (D) All of these

HD0070

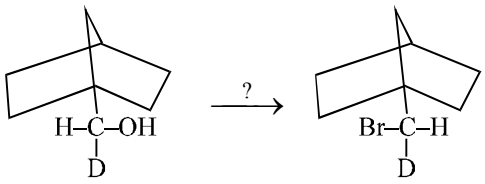
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Al-&-Et-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65"

71.  (A). The product A will be :

(A)  (B)  (C) $H_2C = CH_2$ (D) 

HD0071

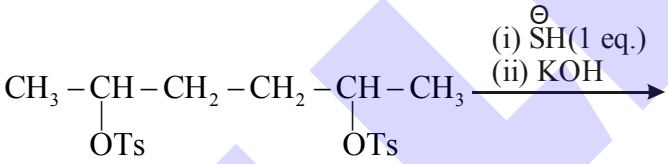
72. Which reaction conditions (reagents) is suitable for the following reaction:

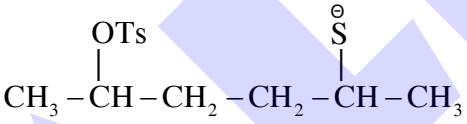
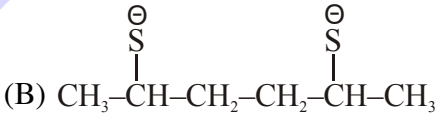


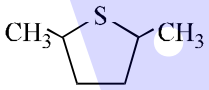
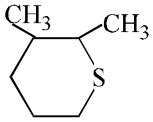
(A) Br_2 / CCl_4 (B) $SOBr_2$ (C) PBr_3 (D) HBr / conc H_2SO_4

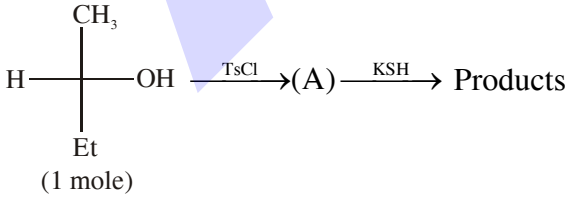
HD0072

73. In the given reaction

 (X), X will be :

(A)  (B) 

(C)  (D)  **HD0073**

74. 

(Assuming all the substrate convert into substitution products containing 0.05 mole of S-configuration)Calculate the percentage of S_N2 mechansim.

(A) 90% (B) 80% (C) 70% (D) 95% **HD0074**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & Et - Hydr - Carb. - Oxi - Red - Ald & Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtrO-I.p65"

75. The reaction of $SOCl_2$ on alkanols to form alkyl chlorides gives good yields because

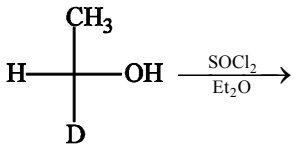
(A) Alkyl chlorides are immiscible with $SOCl_2$

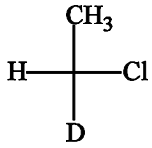
(B) The other products of the reaction are gaseous and escape out

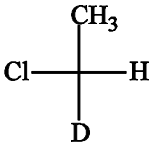
(C) Alcohol and $SOCl_2$ are soluble in water

(D) The reaction does not occurs via intermediate formation of an alkyl chloro sulphite.

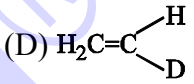
HD0075

76.  (A). The product A will be :

(A) 

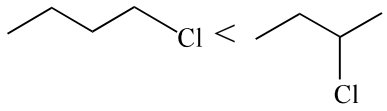
(B) 

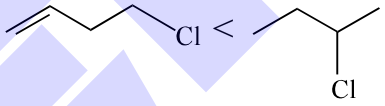
(C) $H_2C = CH_2$

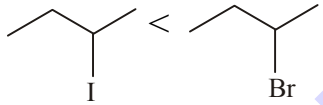
(D) 

HD0076

77. In the given pairs, which pair represent correct order of rate dehydrohalogenation reaction.

(A) 

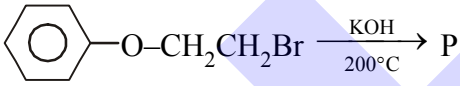
(B) 

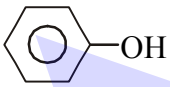
(C) 

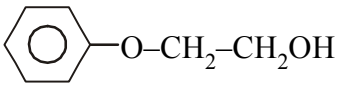
(D) $CH_3 - CH_2 - Cl < CD_3 - CD_2 - Cl$

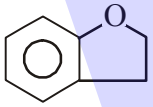
HD0077

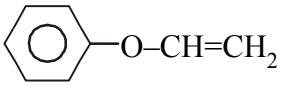
78. The product of the reaction



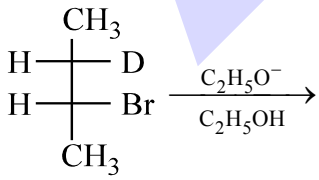
(A) 

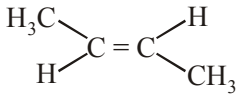
(B) 

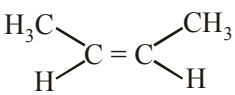
(C) 

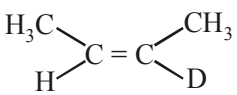
(D) 

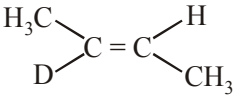
HD0078

79. 

(A) 

(B) 

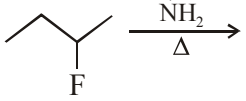
(C) 

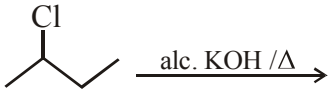
(D) 

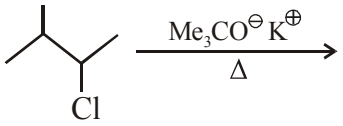
HD0079

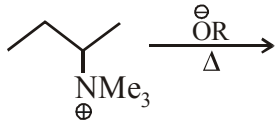
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Halo-Alc & Eth -Hydr -Carb -Oxi - Red - Ald & Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-1.p65"

80. Which of following reaction(s) produce Saytzeff product as a major product :

(A) 

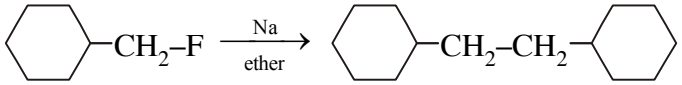
(B) 

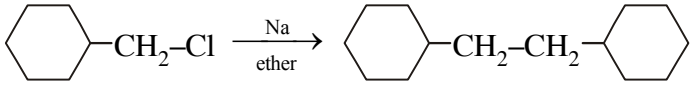
(C) 

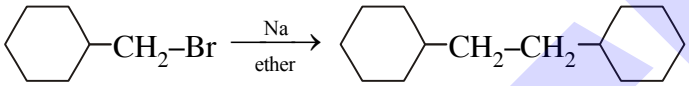
(D) 

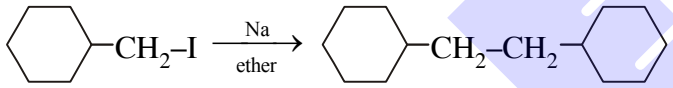
HD0080

81. The correct order of rate of following Wurtz recations :

(I) 

(II) 

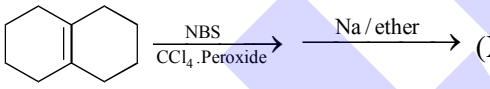
(III) 

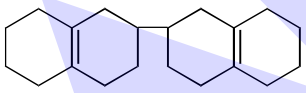
(IV) 

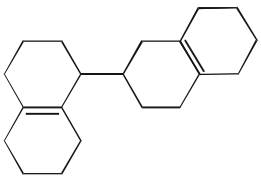
(A) I > II > III > IV (B) II > I > III > IV

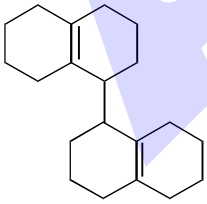
(C) IV > III > II > I (D) In all rate of Wurtz reaction is same

HD0081

82.  (X) ; X is

(A) 

(B) 

(C) 

(D) None of these

HD0082

83. Find out the correct order of rate of reaction towards free radical allylic substitution :

(I) $CH_3–CH=CH_2$ (II) $CH_3–CH_2–CH=CH_2$ (III) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH=CH_2$

(A) I > II > III (B) II > I > III (C) III > II > I (D) III > I > II

HD0083

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & Et.-Hydr.-Garb.-Oxi.-Red.-Akl & Ken-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65

84. What will be the major product, when 2-methyl butane undergoes bromination in presence of light ?

(A) 1-Bromo-2-methyl butane (B) 2-Bromo-2-methyl butane

(C) 2-Bromo-3-methyl butane (D) 1-Bromo-3-methyl butane **HD0084**

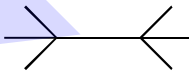
85. Which can not be the possible product of the given reaction ?

$$CH_3-CH_2-\underset{\underset{O}{\|}}{C}-OAg \xrightarrow[CCl_4,\Delta]{Br_2} \text{product(s)}$$

(A) $CH_3 - CH_2 - Br$ (B) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_2-CH_3$

(C) $CH_3 - CH_2 - CH_2 - CH_3$ (D) $CH_3 - CH_2 - CH_3$ **HD0085**

86. Choose that alkane which cannot give only one monochloro derivative upon reaction with chlorine in sun light :

(A)  (B)  (C)  (D) 

HD0086

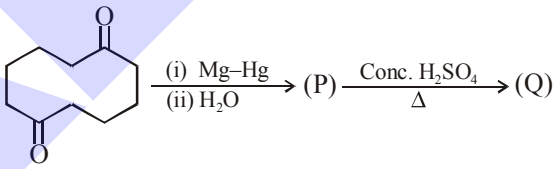
87. 2-chloropentane on halogenation with chlorine gives 2,3, dichloropentane. What will be the structure of free radical species formed in the reaction ?

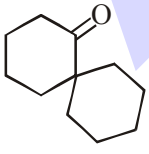
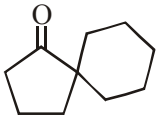
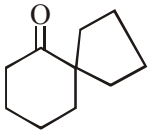
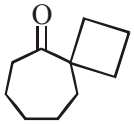
(A) Tetrahedral (B) Trigonal planar (C) Square planar (D) Pyramidal **HD0087**

88. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be -

(A) neopentane (B) propane (C) pentane (D) isopentane **HD0088**

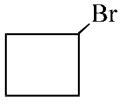
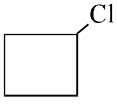
89. Major product (Q) of following reaction is :



(A)  (B)  (C)  (D) 

HD0089

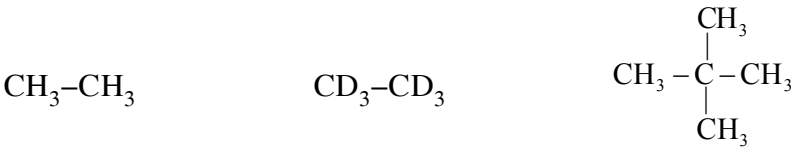
90. 1-Bromo-3-chloro cyclobutane on reaction with 2-equivalent of sodium in ether gives

(A)  (B)  (C)  (D) 

HD0090

node06\B0BC-BD\Kota\JEE(Advanced)\Enfuse\Chem\Module\Mod.#Hdc-Alc.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\01_ExtO-I.p65"

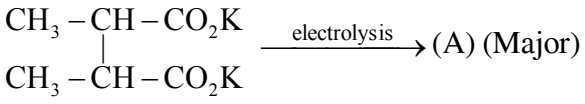
91. Correct order of rate of photochlorination for following compounds is :



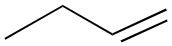
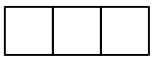
(I) (II) (III)

(A) II < I < III (B) I < II < III (C) III < I < II (D) II < III < I

HD0091

92. 

Major product (A) of above reaction :

(A)  (B)  (C)  (D) 

HD0092

93. During the preparation of ethane by Kolbe's electrolytic method using inert electrode the pH of the electerolyte

(A) Decreases progressively as the reaction proceeds

(B) Increaes progressively as the reaction procces

(C) Remains constant throughout the reaction

(D) May decrease if concentration of the electrolytes is not very high **HD0093**

94. When isobutane is chlorinated in the presence of diffused sunlight, then the product formed is :

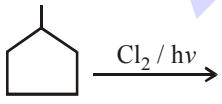
(A) Tertiary butyl chloride in major amount

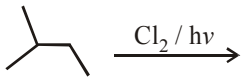
(B) Isobutyl chloride in major amount

(C) Both 50 % each

(D) n-Butyl chloride, isobutyl chloride and sec-butyl chloride are formed **HD0094**

95. Consider the following reactions :

 Total number of monochlorinated product = X (Excluding stereoisomers)

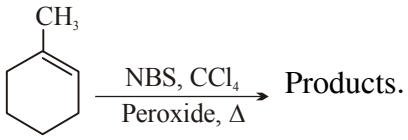
 Total number of monochlorinated product = Y (Excluding stereoisomers)

Identify value of X + Y.

(A) 8 (B) 9 (C) 11 (D) 10 **HD0095**

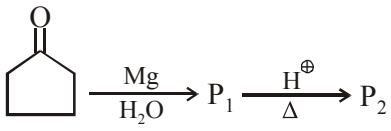
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod#Halo-Alc & Et - Hydr - Carb.-Oxi. Red.-Add & Rxn-(OC)\Eng\01_Halogen Derivatives\01_ExrO-I.p65"

96. Find out the total no. of products (including stereo) in the given reaction :



(A) 8 (B) 9 (C) 10 (D) 11 **HD0096**

97. Which of the following is not correct about P_2 :



(A) It is a spiro compound (B) It is a Ketone

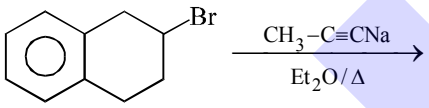
(C) It can show tautomerism (D) Its double bond equivalent is 4 **HD0097**

98. On heating glycerol with excess amount to HI, the product formed is–

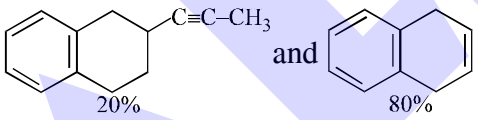
(A) Allyl iodide (B) Isopropyl iodide

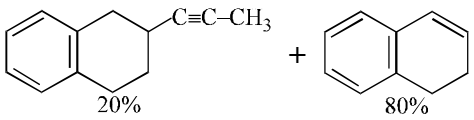
(C) Propylene (D)1,2,3–tri-iodopropane **HD0098**

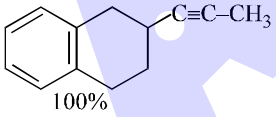
99. In the given reaction:

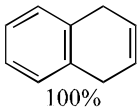


the products are:

(A) 

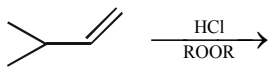
(B) 

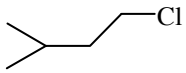
(C) 

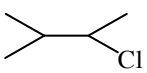
(D) 

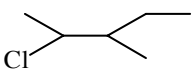
HD0099

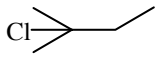
100. Major product of the reaction -



(A) 

(B) 

(C) 

(D) 

HD0100

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EXERCISE # O-2

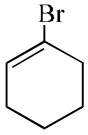
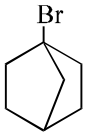
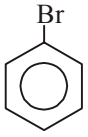
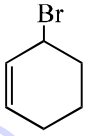
Choose the correct option. One or more than one are correct

1. Rate of S_N2 depends on :

(A) Conc of Nucleophile (B) Conc of substrate

(C) Nature of leaving group (D) Nature of solvent **HD0101**

2. S_N2 reaction will be negligible in

(A)  (B)  (C)  (D) 

HD0102

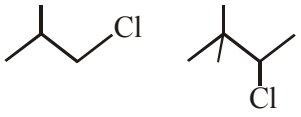
3. $Br-\underset{\underset{CH_3}{|}}{CH}-\overset{14}{C}H=CH_2 \xrightarrow{HI}$

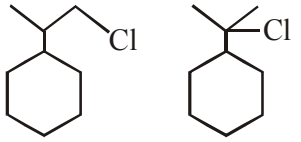
Products which can be obtained during the reaction in good yield :

(A) $Br-\underset{\underset{CH_3}{|}}{CH}-\overset{14}{\underset{\underset{CH_3}{|}}{CH}}-I$ (B) $I-\underset{\underset{CH_3}{|}}{CH}-\overset{14}{\underset{\underset{CH_3}{|}}{CH}}-Br$

(C) $Br-\underset{\underset{CH_3}{|}}{CH}-\overset{14}{C}H_2-CH_2-I$ (D) $I-\underset{\underset{CH_3}{|}}{CH}-CH_2-\overset{14}{C}H_2-Br$ **HD0103**

4. In the given pair in which pair the first compound is more reactive than second towards S_N2 reaction.

(A)  (B) 

(C)  (D)  **HD0104**

5. Consider the given reaction

$$H-\underset{\underset{C_2H_5}{|}}{\overset{\overset{CH_3}{|}}{C}}-OTs \ (S) \xrightarrow[(DMF)]{NaCN} CH_3CH_2\underset{\underset{CH_3}{|}}{C}H-CN$$

which of following statements is/are correct for the above reaction.

(A) Product formation takes place due to the breaking of O–Ts

(B) The reaction is S_N2

(C) The reaction is S_N1

(D) Configuration of product is (R) **HD0105**

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6. Which of the following statements is / are true?

(A) $CH_3–CH_2–CH_2–I$ will react more readily than $(CH_3)_2$ CHI for S_N2 reactions.

(B) $CH_3–CH_2–CH_2–Cl$ will react more readily than $CH_3–CH_2–CH_2–Br$ for S_N2 reaction.

(C) $CH_3–CH_2–CH_2–CH_2–Br$ will react more readily than $(CH_3)_3C–CH_2–Br$ for S_N2 reactions

(D) $CH_3–O–C_6H_4–$ CH_2Br will react more readily than $NO_2–C_6H_5–CH_2Br$ for S_N2 reaction

HD0106

7. Incorrect statement about alkyl halides is / are:

(A) Tertiary alkyl halides undergo S_N2 substitutions

(B) Alkyl iodides on exposure to sunlight gradually darken

(C) Photo iodination is irreversible in presence of HIO_3

(D) A nucleophilic substitution is most difficult in alkyl iodides **HD0107**

8. S_N1 & S_N2 is not favourable in

(A) $H_2C = CH–Cl$ (B) $Ph–CH_2–Cl$ (C) $Ph–Cl$ (D) $H_2C=CH–CH_2–Cl$

HD0108

9. Correct statement(s) for the product(s) of following reaction.

$$CH_2 = CH - CH_2 - Ph \xrightarrow[Cl_2\,/\,500°C]{}$$

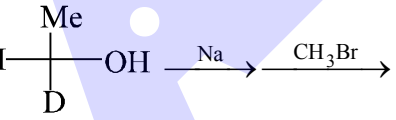
(A) Four different products are formed

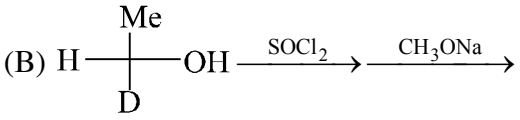
(B) Two optically active products are formed

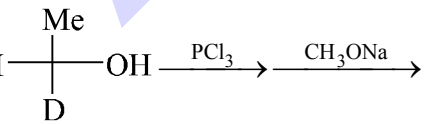
(C) The optically active compound formed here can also be made by the reaction of HCl

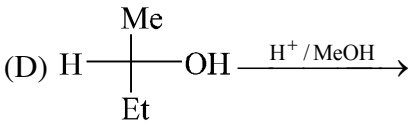
(D) The reaction path is free radical substitution. **HD0109**

10. In which of the following reaction configuration about chiral C is retained in the final product

(A) 

(B) 

(C) 

(D) 

HD0110

11. A gem dichloride is formed in the reaction :

(A) CH_3CHO and PCl_5 (B) CH_3COCH_3 and PCl_5

(C) $CH_2 = CH_2$ and Cl_2 (D) CH_2 = CHCl and HCl **HD0111**

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12. In which product formation takes place according to Hoffmann's rule

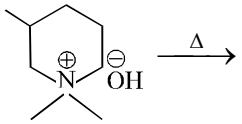
(A) $CH_3CH_2-\underset{\underset{Br}{|}}{CH}-CH_3 \xrightarrow[\Delta]{t\text{-}BuO^{\ominus}K^{\oplus}}$

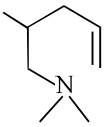
(B) $CH_3CH_2-\underset{\underset{Br}{|}}{CH}-CH_3 \xrightarrow[\Delta]{CH_3CH_2O^{\ominus}K^{\oplus}}$

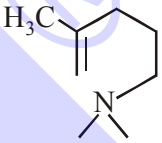
(C) $CH_3CH_2-\underset{\underset{CH_3}{|}}{CH}-\overset{\oplus}{N}(CH_3)_2(CH_3) \xrightarrow[\Delta]{OH^{\ominus}}$

(D) $CH_3CH_2\underset{\underset{\oplus S(CH_3)_2}{|}}{CH}-CH_3 \xrightarrow[\Delta]{OH^{\ominus}}$ **HD0112**

13. Which of following are correct for given reaction



(A) Major product of reaction is 

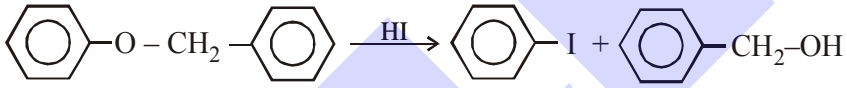
(B) Major product is 

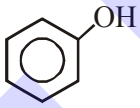
(C) The reaction is thermal elimination reaction (D) The reaction is E_2 reaction **HD0113**

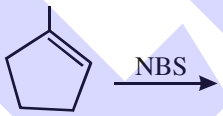
14. In which case incorrect products are formed :

(A) $Me_3C-O-CH_3 \xrightarrow{HI} Me_3C-OH + CH_3I$

(B) $H_3C-O-CH_2-CH_3 \xrightarrow{HI} CH_3OH + ICH_2CH_3$

(C) 

(D)Anisole $\xrightarrow{HI}$  $+ CH_3I$ **HD0114**

15. In the given reaction : 

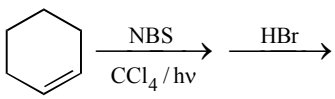
Find out the correct statement

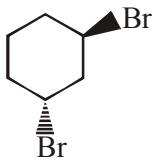
(A) It gives total 9 allylic brominated products

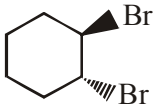
(B) 6 fractions are obtained on fractional distillation of product mixture

(C) Subtrate has 7 allylic hydrogens

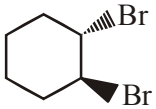
(D) NBS is a brominating agent for allylic positions **HD0115**

16.  (X) + (Y) enantiomeric pair :

(A) 

(B) 

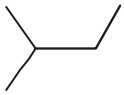
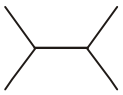
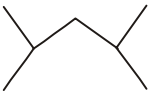
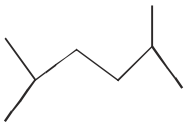
(C) 

(D) 

HD0116

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17. Which of the following can be produced by Wurtz reaction in good yield :

(A)  (B)  (C)  (D) 

HD0117

18. Products formed when HCl adds to 2,4-hexadiene is :

(A) 4-chloro-2-hexene (B) 2-chloro-3-hexene

(C) 2-chloro-4-hexene (D) 1-chloro-2-hexene **HD0118**

19. Correct statement among the following is/are :

(A) The rate of hydrolysis of tertiary butyl bromide increases by addition of Ag_2O

(B) Aqueous Ag_2O produces nucleophilic OH^-

(C) The addition of a small amount of oxygen slows down the photochemical chlorination of methane.

(D) CH_3CH_2Cl is more reactive than $PhCH_2Cl$ for bimolecular nucleophilic substitution reaction

HD0119

20. Incorrect statement among the following is/are :

(A) R—OH with NaI in the presence of phosphoric acid gives R—I, but not in the absence of phosphoric acid

(B) 2-methyl propane on chlorination (Cl_2, hν) gives 1-chloro-2-methyl propane while bromination (Br_2, hν) gives 2-bromo-2-methyl propane

(C) Usually higher temperature prefers substitution over elimination

(D) Triphenyl chloromethane cannot be hydrolysed **HD0120**

21. Correct statements among the following is/are :

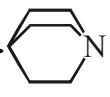
(A) Dihaloalkanes having the same type of halogen atoms on same atom are named as alkylidene dihalides

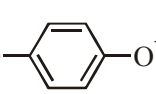
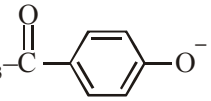
(B) Dihaloalkanes having the same type of halogen atoms on adjacent atoms are named as alkylene dihalides

(C) In common name system gem-dihalides are named as alkylidene halide

(D) In common name system vic-dihalides are named as alkylene halide **HD0121**

22. Which of the following is correct order of nucleophilicity ?

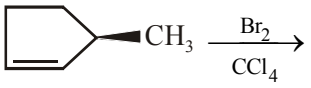
(A) $(CH_3CH_2)_3N >$  (B) HOO– > HO– in DMSO

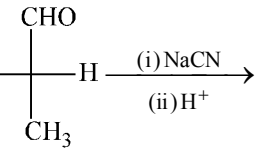
(C) $H_2S > H_2O$ (D) CH_3O–  > CH_3– 

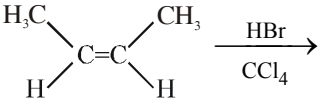
HD0122

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Al- & Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\02_Eff-O-II.p65"

23. Which of following reaction products are diastereomer of each other :

(A) 

(B) 

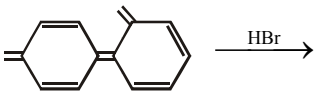
(C) 

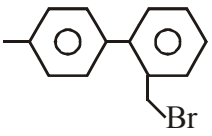
(D) $CH_3-\underset{\underset{Et}{|}}{CH}-CH=CH-Ph \xrightarrow[\text{peroxide}]{HCl}$

(Optically pure)

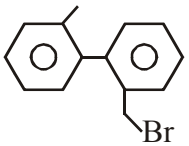
HD0123

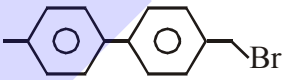
24. Product obtained in given reaction in good yield are :



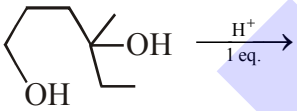
(A) 

(B) 

(C) 

(D) 

HD0124

25. 

Correct statements for given reaction :

(A) Product mixture is resolvable

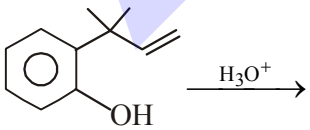
(B) Product can be separated by fractional distillation of mixture

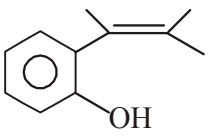
(C) Two products possible & both are optically active

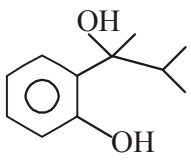
(D) Products are diastereomer

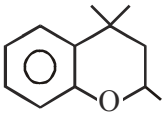
HD0125

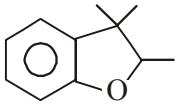
26. Which of the following can be formed during this reaction ?



(A) 

(B) 

(C) 

(D) 

HD0126

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & Et.-Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\01_Halogen Derivatives\02_Et#O-II.p65

27. Select **true** statement(s) :

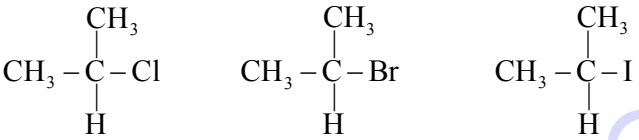
(A) Cyclopropane decolorizes bromine water

(B) In general, bromination is more selective than chlorination.

(C) The 2,4,6-tri-tert, butylphenoxy radical is resistant to dimerization.

(D) The radical-catalysed chlorination, $ArCH_3 \rightarrow ArCH_2Cl$, occurs faster when Ar = phenyl than when Ar = p-nitrophenyl. **HD0127**

28. From left to right, correct statements are :



(A) Rate of S_N1 mechanism increases in polar protic solvent

(B) Rate of S_N2 mechanism increases in DMSO

(C) Rate of E_2 mechanism increases

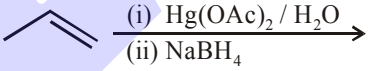
(D) Rate of E_1 mechanism increases **HD0128**

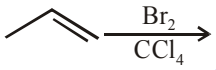
29. Number of following reactions which produces hydrocarbon as major product ?

(i) $CH_3–CH_2–Cl \xrightarrow[Et_2O]{Na}$

(ii) $CH_3–\overset{O}{\overset{\|}{C}}–OK \xrightarrow{Electrolysis}$

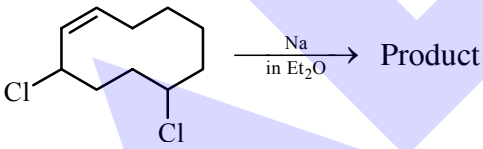
(iii) $\underset{Cl}{\overset{|}{CH_2}}–\underset{Cl}{\overset{|}{CH_2}} \xrightarrow[dust]{Zn}$

(iv) 

(v) 

(vi) $CH_3–CH_2–CH=CH_2 \xrightarrow[(ii)\ CH_3COOH / H_2O]{(i)\ B_2H_6 - THF}$

(A) 2 (B) 4 (C) 5 (D) 6 **HD0129**

30. 

Correct statement is/are :

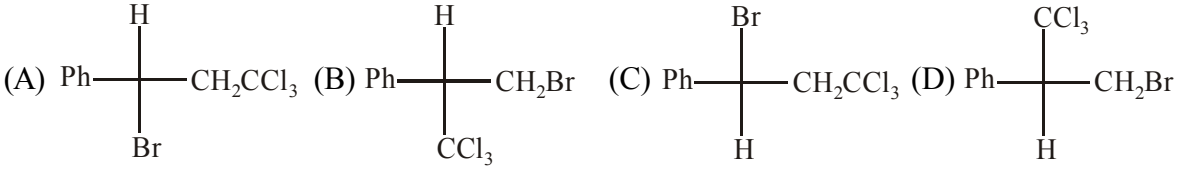
(A) odd no. of double bond equivalent in product

(B) product is bicyclic compound

(C) product can show geometrical isomerism

(D) reaction involve carbocation as intermediate **HD0130**

31. $Ph—CH = CH_2 + BrCCl_3 \xrightarrow{Peroxide}$ Product is :



HD0131

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alk.&Et-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\01_Halogen Derivatives\02_Exe\O-II.p65"

EXERCISE # S-1

Comprehension Type :

Paragraph for Q.No. 01 & 02

Groups like CN & [– O – $\ddot{N}$ = O] possess two nucleophilic centre and are called ambident nucleophiles. Actually cyanide group is hybride of two contributing structures and therefore can act as nucleophile in two different ways [$\overset{\ominus}{C}\equiv N \longleftrightarrow : C = N^{\ominus}$]. Similarly nitrite ion also represents an ambident nucleophile with two different points of linkage [O – $\ddot{N}$ = O].

1. Correct option among the following :

(A) $\underset{\text{Haloalkane}}{R-X} \xrightarrow{KCN} \underset{\text{Major product}}{RNC}$

(B) $R-X \xrightarrow{AgCN} \underset{\text{Major}}{R-CN}$

(C) $R-X \xrightarrow{KNO_2} \underset{\text{Major}}{R-O-N=O}$

(D) $R-X \xrightarrow{AgNO_2} \underset{\text{Major product}}{R-O-N=O}$

HD0132

2. Incorrect statement

$$R-X \xrightarrow[AgCN]{KCN}$$

(A) KCN is predominentely ionic in nature
(B) AgCN is mainly covalent in nature
(C) In AgCN, carbon is the donor atom
(D) In AgCN nitrogen is the donor atom

HD0133

3. **Statement-1 :** HBr shows antimarkownikoff 's addition on propene but not HCl.
Statement-2 : H-Br is stronger acid than H–Cl.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **HD0134**

4. Match the List I with List II and select the correct answer using the codes given below the Lists.

List I	**List II**
(A) $CF_3–CHCl_2 \xrightarrow{alc.KOH/\Delta} CF_2=CCl_2$	(P) Elimination Reaction
(B) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH \xrightarrow{H^{\oplus}} CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=CH_2$	(Q) Carbocation
(C) $CH_3–CH_2–Br \xrightarrow[\Delta]{alc.KOH} CH_2=CH_2$	(R) Carbanion
(D) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle Br}{\overset{|}{\underset{|}{C}}}}-CH_3 \xrightarrow[\Delta]{EtOH} CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=CH_2$	(S) Free radical **HD0135**

node06\B0BC-BD\Kota\JEE(Advanced)\Enfuse\Chem\Module\Word #\Hdo-Alc & Et - Hydr - Garb.-Oxi.-Red.-All & Ket-(OQ)\Eng\01_Hdogen Derivatives \03_Ex#S-1.p65"

5. Match List I with List II and select the correct answer from the codes given below:

List I (Reactions)	**List II (Products)**
(A) $CH_3–O–SO_2CH_3 + C_2H_5\overset{\ominus}{O}$	(P) $CH_3–CH_2–PH_2$
(B) $CH_3–CH_2–I + PH_3$	(Q) $CH_3–O–C_2H_5$
(C) $HC\equiv\overset{\ominus}{C}\overset{\oplus}{N}a + CH_3–CH_2–Br$	(R) $CH_3–O–CH_3$
(D) $CH_3–Cl + CH_3–\overset{\ominus}{O}$	(S) $CH\equiv C–CH_2–CH_3$

HD0136

6. Match List-I with List-II for given S_N2 reaction & select the correct answer from the codes given below

$$Z–CH_2Br + CH_3O^{\ominus} \longrightarrow Z–CH_2–OCH_3 + Br^{\ominus}$$

List-I (Z–)	**List-II (relative reactivity)**
(A) H–	(P) 0.1
(B) $CH_3–$	(Q) 3
(C) $C_2H_5–$	(R) 1
(D) $(CH_3)_2CH–$	(S) 100

HD0137

7. Match the List I with List II and select the correct answer using the codes given below the Lists.

List I	**List II**
(A) E_{1CB}	(P) $CH_3CH_2CH_2\overset{\oplus}{N}(CH_3)_2–O^{\ominus}$
(B) Saytzeff alkene as major product	(Q) $CH_3CH_2CH_2–O–\overset{\overset{S}{\|\|}}{C}–S–CH_3$
(C) E_2	(R) $CH_3–CH_2–\underset{\underset{Cl}{\|}}{CH}–CH_3$
(D) E_i	(S) $C_6H_5–CH_2–\underset{\underset{F}{\|}}{CH}–CH_3$

HD0138

8.

Column - I (Reactions)	**Column - II (Characteristics)**
(A) $CH_3CH_2CH=CH_2 \xrightarrow{HBr}$	(P) Bimolecular
(B) $CH_3CH_2CH=CH_2 \xrightarrow{HBr,\ Peroxide}$	(Q) Carbocation intermediate
(C) $PhCH(CH_3)OH \xrightarrow{SOCl_2}$	(R) Regioselective
(D) $PhCH(CH_3)OH \xrightarrow{HBr}$	(S) Racemic modification
	(T) Stereospecific reaction

HD0139

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol.#\Hdb-Al,&,Et-Hydr.-Carb.-Oxi.-Red.-All&Ram (OC)\Eng\01_Halogen Derivative\03_Ex#5-1.p65"

9. **Column - I** **Column - II**

(Statements) **(Consistent path of reaction)**

(A) Reactions are concerted (P) S_N1

(B) CH_3X cannot react (Q) S_N2

(C) 3° R–X > 2° R–X > 1° R–X (R) E1

(D) R–I reacts faster than R–Cl (S) E2 **HD0140**

10. Each of the compounds in column A is subjected to further chlorination. **Match the following** for them :

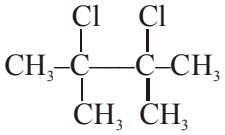
Column – A **Column – B**

(A) $CHCl_2–CH_2–CH_3$ (P) Optically active original compound

(B) $CH_2Cl–CHCl–CH_3$ (Q) Only one trichloro product

(C) $CH_2Cl–CH_2–CH_2–Cl$ (R) Three trichloro product.

(D) $CH_3–CCl_2–CH_3$ (S) Four trichloro product

(E)  (T) Atleast one of the trichloro product is optically active

(U) Two trichloro products. **HD0141**

11. **Column - I** **Column - II**

(Intermediate)

(A) Carbocation (P) Kolbey Electrolysis

(B) Carbanion (Q) Wurtz reaction

(C) Free radical (R) Dehydration of alcohol

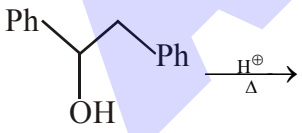
(D) Octet complete in one of the intermediate (S) Monocarboxylic acid with sodalime

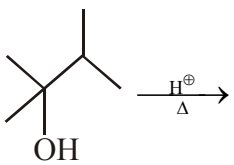
HD0142

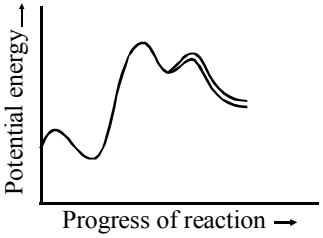
12. Match the column

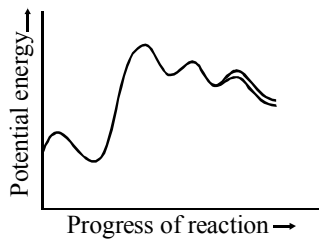
Column–I **Column–II**

(Reaction) **(Potential energy curve)**

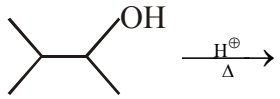
(A) 

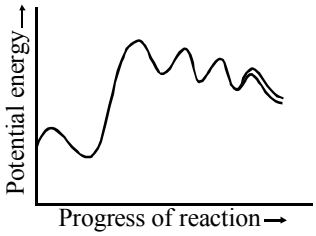
(B) 

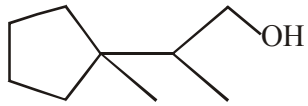
(P) 

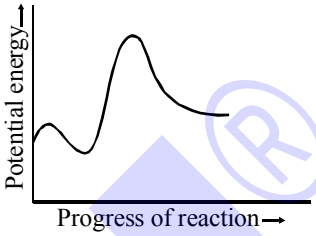
(Q) 

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Al.c & El.-Hydr.-Garb.-Oxi.-Red.-All&Rom-(OQ)\Eng\01_Halogen Derivatives\03_Ex#5-1.p65"

(C) 

(R) 

(D) 

(S) 

HD0143

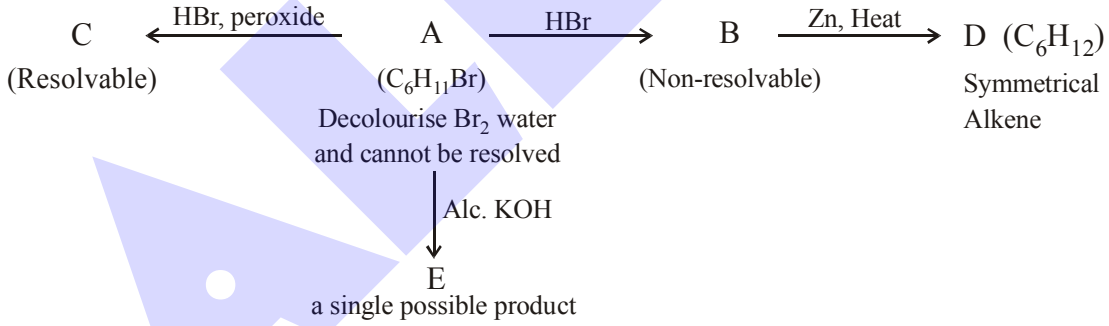
Subjective Type :

13. RCl is treated with Li in ether to form R – Li, R – Li reacts with water to form isopentane. R – Cl also reacts with sodium to form 2, 7–dimethyloctane. What is the structure of R – Cl.

HD0144

14. A chloroderivative 'X' on reduction gave a hydrocarbon with five carbon atoms in the molecule. When X is dissolved in ether and treated with sodium, 2, 2, 5, 5-tetramethyl hexane is obtained. What is compound X. **HD0145**

15.



Identify A, C & E in the sequence of reaction. **HD0146**

16. With the help of following data show HBr exhibits the peroxide effect.

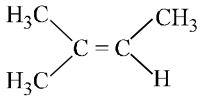
	ΔH_1^0/kJ mol^{-1}	ΔH_2^0/kJ mol^{-1}
H–X	$\dot{X} + CH_2 = CH_2 \rightarrow X\,CH_2 - \dot{C}H_2$	$XCH_2-\dot{C}H_2 + H-X \rightarrow XCH_2CH_3 + \dot{X}$
HCl	–67	+ 12.6
HBr	–25.1	– 50.2
HI	+46	–117.1

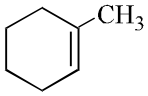
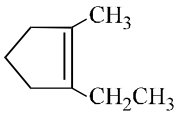
HD0147

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod#Hdb-Alc.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivative\03_Exr#S-I.p65"

17. Write all the monochlorinated products (including stereo) of isohexane. **HD0148**

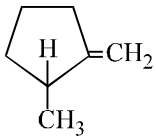
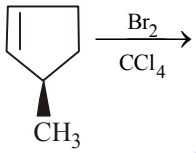
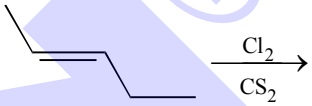
18. What are the products of the following reactions ?

(a) $PhCH = CHCH_3 + HBr \longrightarrow A$ (b)  $+ HI \longrightarrow B$

(c)  $+ HBr \xrightarrow{\text{Peroxide}} C$ (d)  $+ HCl \longrightarrow D$ **HD0149**

19. It required 0.7 g of a hydrocarbon (A) to react completely with Br_2 (2.0 g) and form a non resolvable product. On treatment of (A) with HBr it yielded monobromo alkane (B). The same compound (B) was obtained when (A) was treated with HBr in presence of peroxide. Write down the structure formula of (A) and (B) and explain the reactions involved. **HD0150**

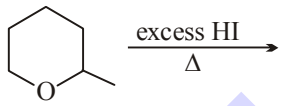
20. Complete following reaction :

(a)  (b)  (c)  **HD0151**

21. $CH_3–CH_2I$ reacts more rapidly with strong base in comparison to CD_3CH_2I. **HD0152**

22. $CH\equiv C–CH_2–CH=CH_2$, adds up HBr to give $CH\equiv C–CH_2–CHBr–CH_3$ while $CH\equiv C–CH=CH_2$ adds up HBr to give $CH_2 = \underset{\displaystyle Br}{\underset{|}{C}} - CH = CH_2$ **HD0153**

23. Predict the product(s) and write the mechanism of the given reaction :

 **HD0154**

24. What are the products of the following reactions?

(a) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-Cl + \overline{O}CH_3 \longrightarrow$ (b) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-O^- + CH_3 - X \longrightarrow$ **HD0155**

25. A primary alkyl bromide (A), C_4H_9Br, reacted with alcoholic KOH to give compound (B). Compound (B) reacted with HBr to give an isomer of (A). When (A) was reacted with sodium metal it gave compound (D), C_8H_{18}, which was different from the compound produced when n-butyl bromide was reacted with sodium. Draw the structure of (A) and write equations for all the reactions.

HD0156

26. In study of chlorination of propane four products (A,B,C,D) of molecular formula $C_3H_6Cl_2$ were obtained. On further chlorination of the above products A gave one trichloro product, B gave two whereas C and D gave three each. When optically active C was chlorinated one of trichloro propanes was optically active and remaining two were optically inactive. Identify the structures of A,BC and D, and explain formation of products. **HD0157**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol.#\Halo-Al-c & El.-Hydr.-Garb.-Oxi.-Red.-All&Rom-(OQ)\Eng\01_Hydrogen Derivatives \03_Ex#5-1.p65"

EXERCISE # J-MAINS

1. Following reaction :

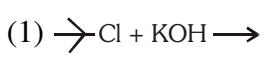
$$(CH_3)_3C–Br + H_2O \longrightarrow (CH_3)_3C–OH + HBr$$

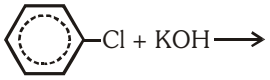
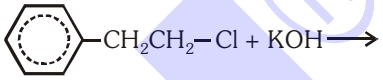
is an example of- **[AIEEE-2002]**

(1) Elimination reaction (2) Free radical substitution

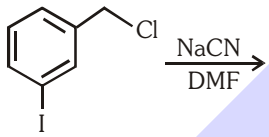
(3) Nucleophilic substitution (4) Electrophilic substitution **HD0158**

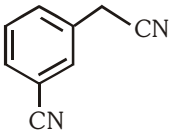
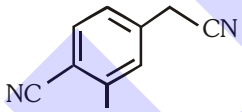
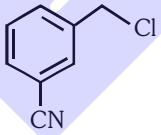
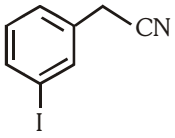
2. SN^1 reaction is feasible in- **[AIEEE-2002]**

(1)  (2) 

(3)  (4)  **HD0159**

3. The structure of the major product formed in the following reaction is : **[AIEEE-2006]**



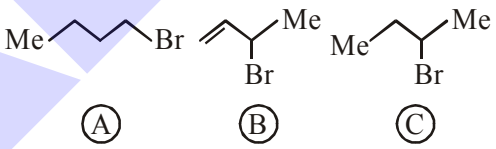
(1)  (2)  (3)  (4)  **HD0161**

4. Which of the following on heating with aqueous KOH, produces acetaldehyde ? **[AIEEE-2009]**

(1) CH_2ClCH_2Cl (2) CH_3CHCl_2 (3) CH_3COCl (4) CH_3CH_2Cl

HD0162

5. Consider the following bromides :- **[AIEEE-2010]**



The correct order of S_N^1 reactivity is

(1) A > B > C (2) B > C > A (3) B > A > C (4) C > B > A **HD0163**

6. In S_N2 reactions, the correct order of reactivity for the following compounds : **[JEE(Main)-2014]**

CH_3Cl, CH_3CH_2Cl, $(CH_3)_2CHCl$ and $(CH_3)_3CCl$ is :

(1) $CH_3CH_2Cl > CH_3Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

(2) $(CH_3)_2CHCl > CH_3CH_2Cl > CH_3Cl > (CH_3)_3CCl$

(3) $CH_3Cl > (CH_3)_2CHCl > CH_3CH_2Cl > (CH_3)_3CCl$

(4) $CH_3Cl > CH_3CH_2Cl > (CH_3)_2CHCl > (CH_3)_3CCl$ **HD0164**

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7. In a nucleophilic substitution reaction : **[JEE(Main)-On-Line-2014]**

$R - Br + Cl^- \xrightarrow{DMF} R - Cl + Br^-$,

which one of the following undergoes complete inversion of configuration?

(1) $C_6H_5CCH_3C_6H_5Br$ (2) $C_6H_5CHCH_3Br$

(3) $C_6H_5CHC_6H_5Br$ (4) $C_6H_5CH_2Br$ **HD0165**

8. The major product obtained in the photo catalysed bromination of 2-methylbutane is :-

(1) 2-bromo-2-methylbutane **[JEE(Main)-On-Line-2014]**

(2) 2-bromo-3-methylbutane

(3) l-bromo-2-methylbutane

(4) l-bromo-3-methylbutane **HD0166**

9. In the presence of peroxide, HCℓ and HI do not give anti-Markownikoff s addition to alkenes because :- **[JEE(Main)-On-Line-2014]**

(1) All the steps are exothermic in HCl and HI

(2) One of the steps is endothermic in HCl and HI

(3) HCl is oxidizing and the HI is reducing

(4) Both HCl and HI are strong acids **HD0167**

10. The synthesis of alkyl fluoride is best accomplished by : **[JEE(Main)-2015]**

(1) Finkelstein reaction (2) Swarts reaction

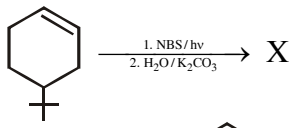
(3) Free radical fluorination (4) Sandmeyer's reaction **HD0169**

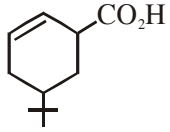
11. 2-chloro-2-methylpentane on reaction with sodium methoxide in methanol yields :

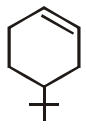
(1) $C_2H_5CH_2\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OCH_3$ (2) $C_2H_5CH_2\underset{\underset{CH_3}{|}}{C}=CH_2$ (3) $C_2H_5CH_2=\underset{\underset{CH_3}{|}}{C}-CH_3$ **[JEE-MAIN-2016]**

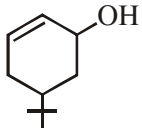
(1) (1) and (2) (2) All of these (3) (1) and (3) (4) (3) only **HD0170**

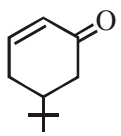
12. The product of the reaction given below is : **[JEE-MAIN-2016]**



(1) 

(2) 

(3) 

(4) 

HD0171

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13. The reaction of propene with HOCl ($Cl_2 + H_2O$) proceeds through the intermediate : **[JEE-MAIN-2016]**

(1) $CH_3–CHCl–CH_2^+$ (2) $CH_3–CH^+–CH_2–OH$

(3) $CH_3–CH^+–CH_2–Cl$ (4) $CH_3–CH(OH)–CH_2^+$ **HD0172**

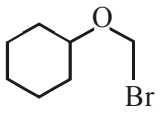
14. The increasing order of the reactivity of the following halides for the S_N1 reaction is : **[JEE-MAIN-2017]**

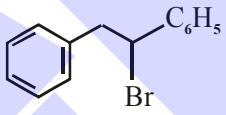
$CH_3\underset{\underset{Cl}{|}}{C}HCH_2CH_3$ (I) $CH_3CH_2CH_2Cl$ (II) p-$H_3CO–C_6H_4–CH_2Cl$ (III)

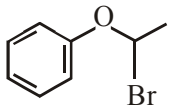
(1) (III) < (II) < (I) (2) (II) < (I) < (III) (3) (I) < (III) < (II) (4) (II) < (III) < (I)

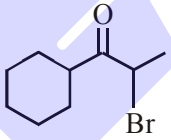
HD0173

15. Which of the following , upon treatment with tert-BuONa followed by addition of bromine water, fails to decolourize the colour of bromine ? **[JEE-MAIN-2017]**

(1) 

(2) 

(3) 

(4) 

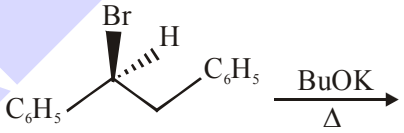
HD0174

16. 3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is : **[JEE-MAIN-2017]**

(1) Six (2) Zero (3) Two (4) Four

HD0175

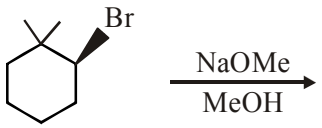
17. The major product obtained in the following reaction is : **[JEE-MAIN-2017]**

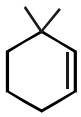


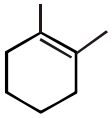
(1) $(\pm)C_6H_5CH(O^tBu)CH_2CH_6H_5$ (2) $C_6H_5CH{=}CHC_6H_5$

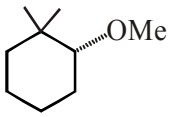
(3) $(+)C_6H_5CH(O^tBu)CH_2H_5$ (4) $(-)C_6H_5CH(O^tBu)CH_2C_6H_5$ **HD0176**

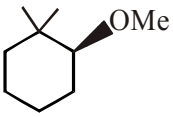
18. The major product of the following reaction is : **[JEE-MAIN-2018]**



(1) 

(2) 

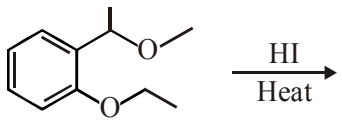
(3) 

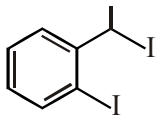
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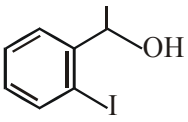
HD0177

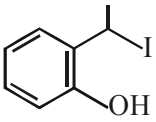
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdc-Al-& Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\04_Ex#Mains.p65

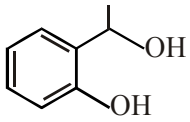
19. The major product formed in the following reaction is : **[JEE-MAIN-2018]**



(1) 

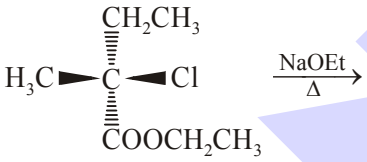
(2) 

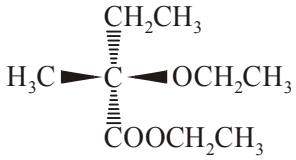
(3) 

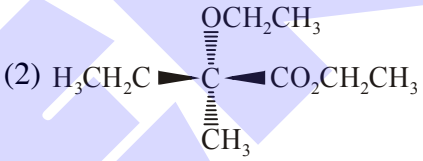
(4) 

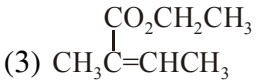
HD0178

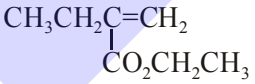
20. The major product of the following reaction is: **[JEE-MAIN-(January) -2019]**



(1) 

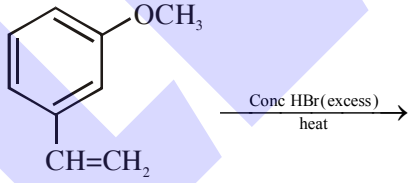
(2) 

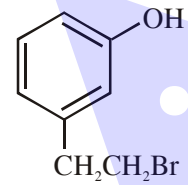
(3) 

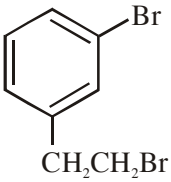
(4) 

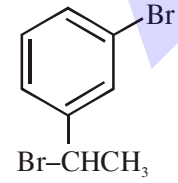
HD0179

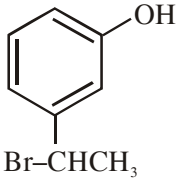
21. The major product of the following reactions: **[JEE-MAIN-(April) -2019]**



(1) 

(2) 

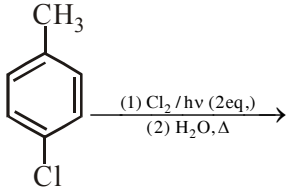
(3) 

(4) 

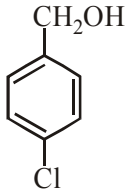
HD0181

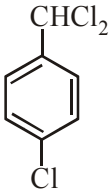
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & Et - Hydr - Carb.-Oxi. Red. Add & Rxn (OC)\Eng\01_Halogen Derivatives\04_Exe\Mains.p65

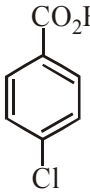
22. The major product of the following reaction is:

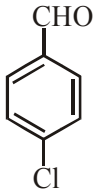


[JEE-MAIN-(April) -2019]

(1) 

(2) 

(3) 

(4) 

HD0182

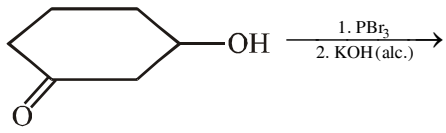
23. Which one of the following alkenes when treated with HCl yields majorly an anti Markovnikov product?

[JEE-MAIN-(April) -2019]

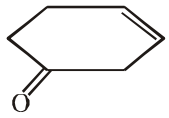
(1) $F_3C - CH = CH_2$ (2) $Cl - CH = CH_2$

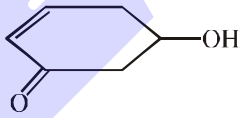
(3) $CH_3O - CH = CH_2$ (4) $H_2N - CH = CH_2$ **HD0183**

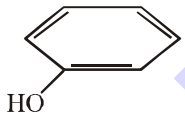
24. The mojor product of the following reaction is :

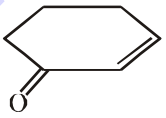


[JEE-MAIN-(April) -2019]

(1) 

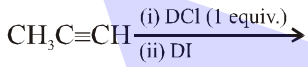
(2) 

(3) 

(4) 

HD0184

25. The mojor product of the following reaction is :



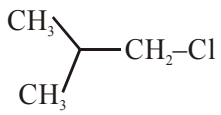
[JEE-MAIN-(April) -2019]

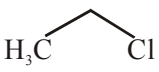
(1) $CH_3CD(Cl)CHD(I)$ (2) $CH_3CD_2CH(Cl)(I)$

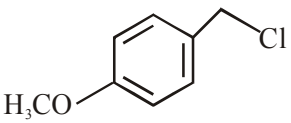
(3) $CH_3CD(I)CHD(Cl)$ (4) $CH_3C(I)(Cl)CHD_2$ **HD0185**

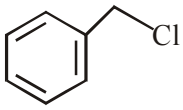
26. Increasing order of reactivity of the following compounds for S_N1 substitution is:

[JEE-MAIN-(April) -2019]

(A) 

(B) 

(C) 

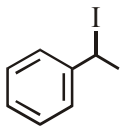
(D) 

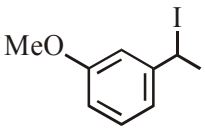
(1) (B) < (C) < (D) < (A) (2) (A) < (B) < (D) < (C)

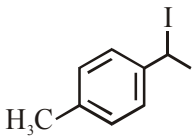
(3) (B) < (A) < (D) < (C) (4) (B) < (C) < (A) < (D) **HD0186**

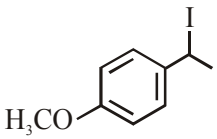
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Al-& Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\04_Ex#Mains.p65

27. Increasing rate of S_N1 reaction in the following compounds is :


(A)


(B)


(C)

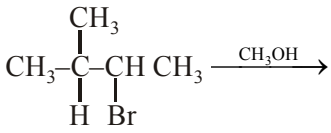

(D)

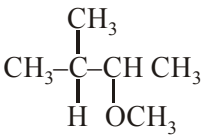
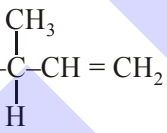
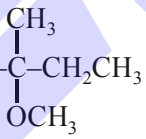
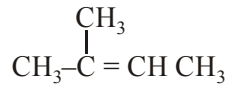
[JEE-MAIN-(April) -2019]

(1) (A) < (B) < (C) < (D)
(2) (B) < (A) < (D) < (C)
(3) (B) < (A) < (C) < (D)
(4) (A) < (B) < (D) < (C)

HD0188

28. The major product of the following reaction is :- **[JEE-MAIN-(April) -2019]**



(1) 
(2) 
(3) 
(4) 

HD0189

29. The increasing order of nucleophilicity of the following nucleophiles is :

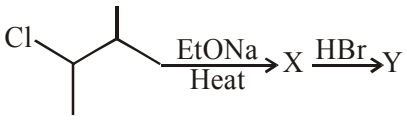
[JEE-MAIN-(April) -2019]

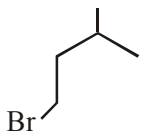
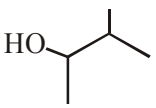
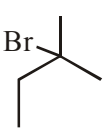
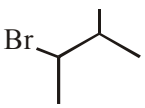
(a) $CH_3CO_2^{\ominus}$ (b) H_2O (c) $CH_3SO_3^{\ominus}$ (d) $\overset{\ominus}{O}H$

(1) (b) < (c) < (a) < (d)
(2) (a) < (d) < (c) < (b)
(3) (d) < (a) < (c) < (b)
(4) (b) < (c) < (d) < (a)

HD0190

30. The major product 'Y' in the following reaction is: **[JEE-MAIN-(April) -2019]**



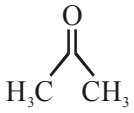
(1) 
(2) 
(3) 
(4) 

HD0191

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & En - Hydr - Carb.-Ox.- Red.- Ald & Ket-(OC)\Eng\01_Halogen Derivatives\04_Ex#Mains.p65

31. The major product of the following addition reaction is : **[JEE-MAIN-(April) -2019]**

$$H_3C - CH = CH_2 \xrightarrow{Cl_2/H_2O}$$

(1) $CH_3 - \underset{Cl}{CH} - \underset{OH}{CH_2}$ (2) $H_3C - \underset{OH}{CH} - \underset{Cl}{CH_2}$ (3) $H_3C -$  (4) 

HD0192

32. The reaction of 2, 4-hexadiene with one equivalent of bromine at 0ºC gives a mixture of two compounds 'X' and 'Y'. If 'X' is 4, 5 - dibromohex-2-ene, 'Y' is - **[NSE -2019]**

(1) 2,5-dibromohex-2-ene (2) 2,5-dibromohex-3-ene

(3) 2,3-dibromohex-3-ene (4) 3,4-dibromohex-3-ene **HD0194**

33. Consider the following reactions : **[JEE-MAIN-2020]**

(a) $(CH_3)_3CCH(OH)CH_3 \xrightarrow{conc.H_2SO_4}$

(b) $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{alc.KOH}$

(c) $(CH_3)_2CHCH(Br)CH_3 \xrightarrow[\text{It should be } (CH_3)_3CO^{\ominus}K^{\oplus}]{\text{given by NTA } (CH_3)_3O^{\ominus}K^{\oplus}}$

(d) $\underset{OH}{(CH_3)_2C}-CH_2-CHO \xrightarrow{\Delta}$

Which of these reaction(s) will not produce Saytzeff product ? **[JEE-MAIN-2020]**

(1) (c) only (2) (a), (c) and (d)

(3) (d) only (4) (b) and (d) **HD0195**

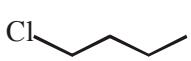
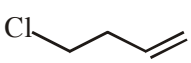
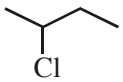
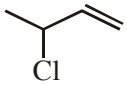
34. Arrange the following bonds according to their average bond energies in descending order :

C–Cl, C–Br, C–F, C–I **[JEE-MAIN-2020]**

(1) C–I > C–Br > C–Cl > C–F (2) C–Br > C–I > C–Cl > C–F

(3) C–F > C–Cl > C–Br > C–I (4) C–Cl > C–Br > C–I > C–F **HD0196**

35. The decreasing order of reactivity towards dehydrohalogenation (E_1) reaction of the following compounds is : **[JEE-MAIN-2020]**

(A)  (B)  (C)  (D) 

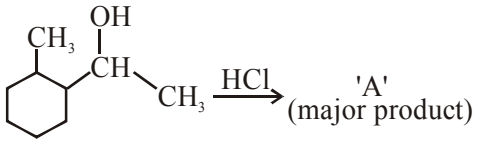
(1) B > D > A > C (2) B > D > C > A

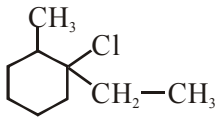
(3) D > B > C > A (4) B > A > D > C

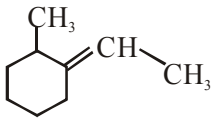
HD0197

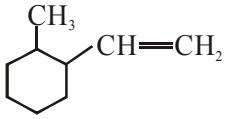
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\04_Exe\Mains.p65"

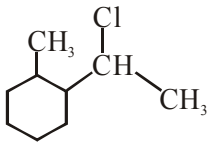
36. What is the final product (major) 'A' in the given reaction ? **[JEE-MAIN-2021]**



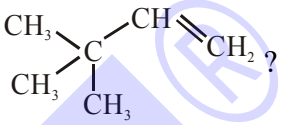
(1) 

(2) 

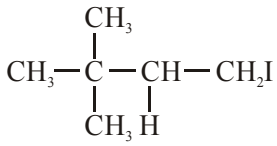
(3) 

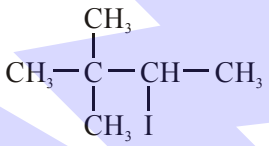
(4) 

HD0228

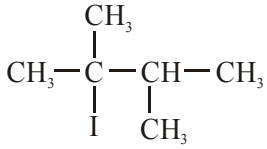
37. What is the major product formed by HI on reaction with 

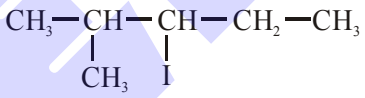
?

(1) 

(2) 

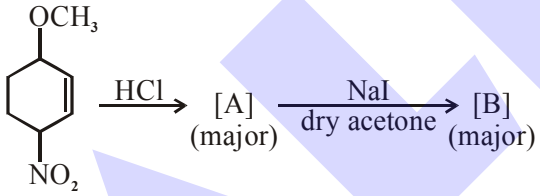
[JEE-MAIN-2021]

(3) 

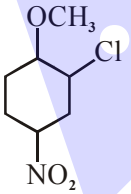
(4) 

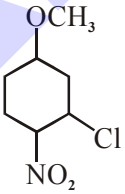
HD0229

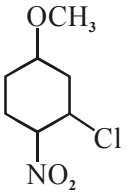
38. Identify A and B in the chemical reaction.

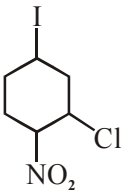


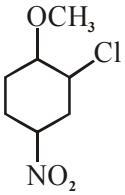
[JEE-MAIN-2021]

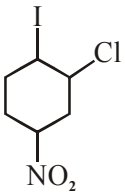
(1) A = 

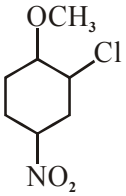
B = 

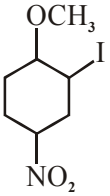
(2) A = 

B = 

(3) A = 

B = 

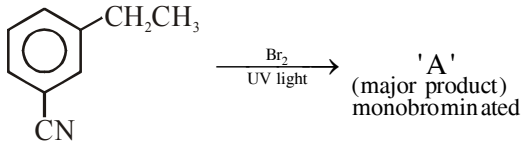
(4) A = 

B = 

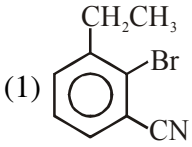
HD0230

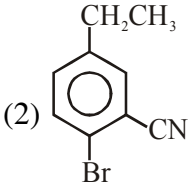
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr - Carb.-Oxi. Red. Add & Rxn (OC)\Eng\01_Halogen Derivatives\04_ExtMains.p65

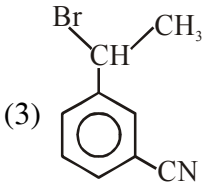
39. For the given reaction : **[JEE-MAIN-2021]**

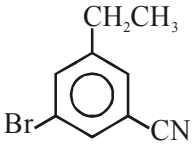


What is 'A'?

(1) 

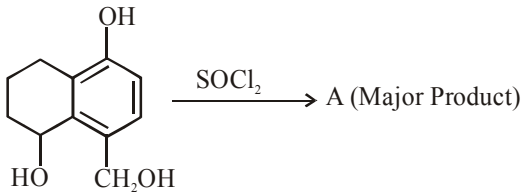
(2) 

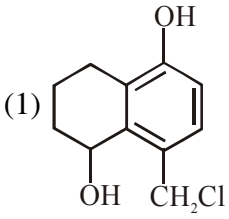
(3) 

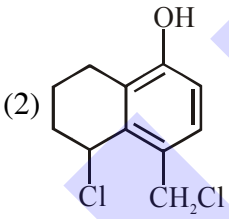
(4) 

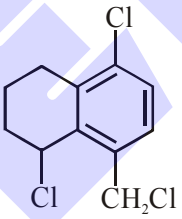
HD0231

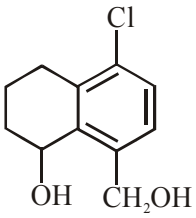
40. Identify A in the given reaction. **[JEE-MAIN-2021]**



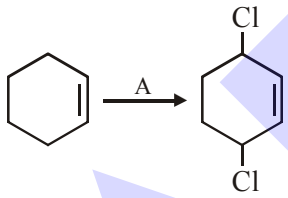
(1) 

(2) 

(3) 

(4) 

HD0232

41. 

[JEE-MAIN-2021]

Identify the reagent(s) 'A' and condition(s) for the reaction :

(1) A = HCl; Anhydrous $AlCl_3$

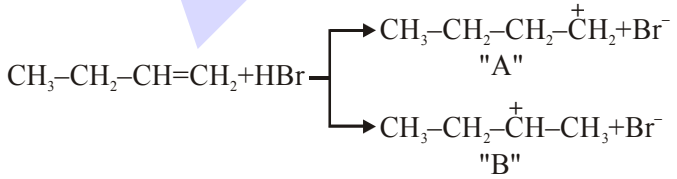
(2) A = HCl, $ZnCl_2$

(3) A = Cl_2; UV light

(4) A = Cl_2; dark, Anhydrous $AlCl_3$

HD0233

42. Choose the correct statement regarding the formation of carbocations A and B given :-



[JEE-MAIN-2021]

(1) Carbocation B is more stable and formed relatively at faster rate

(2) Carbocation A is more stable and formed relatively at slow rate

(3) Carbocation B is more stable and formed relatively at slow rate

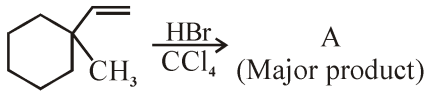
(4) Carbocation A is more stable and formed relatively at faster rate

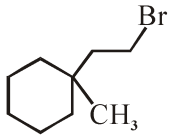
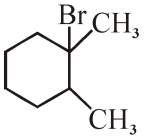
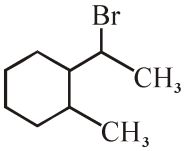
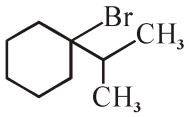
HD0234

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Al-& Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\04_Ex#Mains.p65"

43. The total number of C–C sigma bond/s in mesityl oxide ($C_6H_{10}O$) is_____.
(Round off to the Nearest Integer). **[JEE-MAIN-2021]**

HD0235

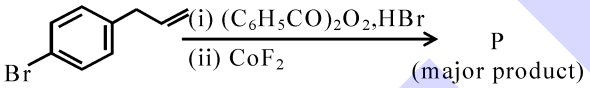
44.  **[JEE-MAIN-2021]**

(1) 
(2) 
(3) 
(4) 

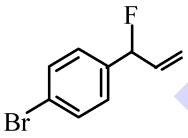
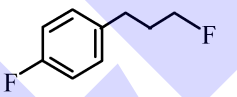
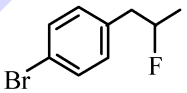
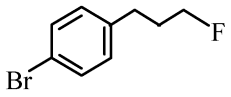
HD0236

45. Given below are two statements :
Statement I : C_2H_5OH and AgCN both can generate nucleophile.
Statement II : KCN and AgCN both will generate nitrile nucleophile with all reaction conditions.
Choose the most appropriate option : **[JEE-MAIN-2021]**
(1) Statement I is true but statement II is false
(2) Both statement I and statement II are true
(3) Statement I is false but statement II is true
(4) Both statement I and statement II are false

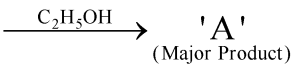
HD0237

46.  **[JEE-MAIN-2021]**

Major product P of above reaction, is :

(1) 
(2) 
(3) 
(4) 

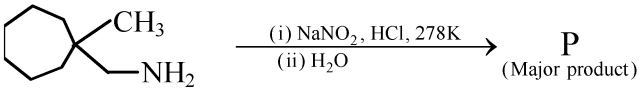
HD0238

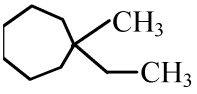
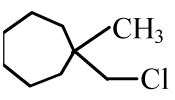
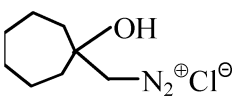
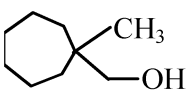
47. In the given reaction 3-Bromo-2, 2-dimethyl butane  Product A is :

(1) 2-Ethoxy-3, 3-dimethyl butane **[JEE-MAIN-2021]**
(2) 1-Ethoxy-3, 3-dimethyl butane
(3) 2-Ethoxy-2, 3-dimethyl butane
(4) 2-Hydroxy-3, 3-dimethyl butane

p **HD0239**

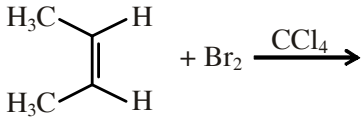
48. What is the major product "P" of the following reaction ? **[JEE-MAIN-2021]**



(1) 
(2) 
(3) 
(4) 

HD0240

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #Halo-Alc & Et- Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\01_Halogen Derivatives\04_Exe#Mains.p65

49. 

Consider the above chemical reaction. The total number of stereoisomers possible for Product 'P' is _____ . **[JEE-MAIN-2021]**

HD0241

50. The stereoisomers that are formed by electrophilic addition of bromine to trans-but-2-ene is/are :
(1) 2 enantiomers and 2 mesomers (2) 2 identical mesomers **[JEE-MAIN-2021]**
(3) 2 enantiomers (4) 1 racemic and 2 enantiomers

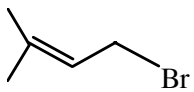
HD0242

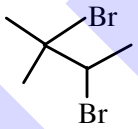
51. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Heavy water is used for the study of reaction mechanism. **[JEE-MAIN-2021]**
Reason (R) : The rate of reaction for the cleavage of O - H bond is slower than that of O-D bond.
Choose the most appropriate answer from the options given below :
(1) Both (A) and (R) are true but (R) is not the true explanation of (A).
(2) Both (A) and (R) are true and (R) is the true explanation of (A).
(3) (A) is false but (R) is true.
(4) (A) is true but (R) is false.

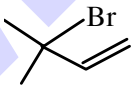
HD0243

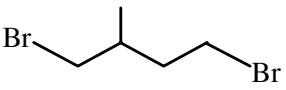
52. The major product formed in the following reaction is : **[JEE-MAIN-2021]**



(1) 

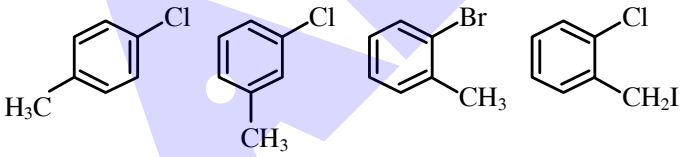
(2) 

(3) 

(4) 

HD0244

53. Among the following compounds I-IV, which one forms a yellow precipitate on reacting sequentially with (i) NaOH (ii) dil. HNO_3 (iii) $AgNO_3$? **[JEE-MAIN-2021]**

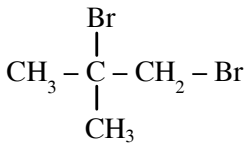


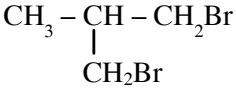
I II III IV

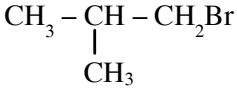
(1) II (2) IV (3) I (4) III

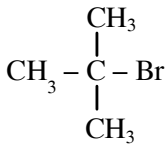
HD0245

54. Excess of isobutane on reaction with Br_2 in presence of light at 125°C gives which one of the following, as the major product? **[JEE-MAIN-2021]**

(1) 

(2) 

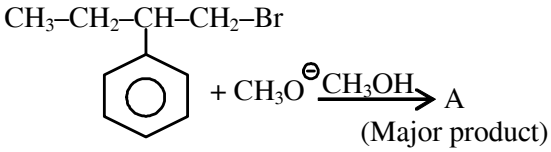
(3) 

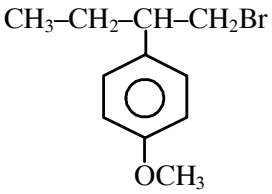
(4) 

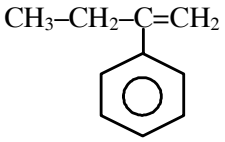
HD0246

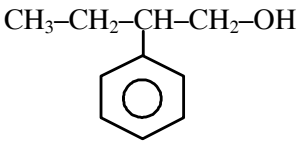
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Al-& Et-Hydr.-Carb.-Oxi.-Red.-Ald.& Ket-(OC)\Eng\01_Halogen Derivatives\04_Ex#Mains.p65"

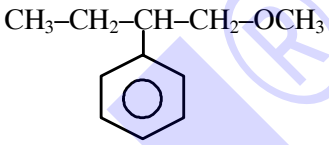
55. The major product (A) formed in the reaction given below is : **[JEE-MAIN-2021]**



(1) 

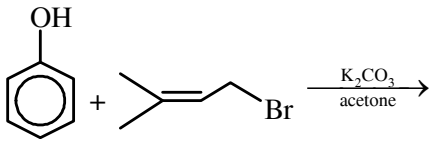
(2) 

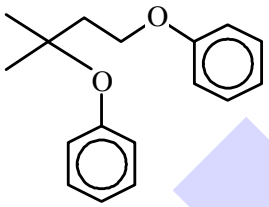
(3) 

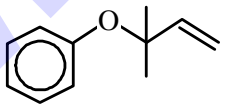
(4) 

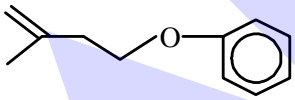
HD0247

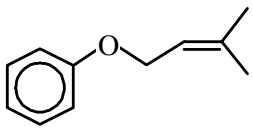
56. The major product of the following reaction, if it occurs by S_N2 mechanism is : **[JEE-MAIN-2021]**



(1) 

(2) 

(3) 

(4) 

HD0248

57. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). **[JEE-MAIN-2021]**

Assertion (A): Synthesis of ethyl phenyl ether may be achieved by Williamson synthesis.

Reason (R): Reaction of bromobenzene with sodium ethoxide yields ethyl phenyl ether.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) (A) is correct but (R) is not correct

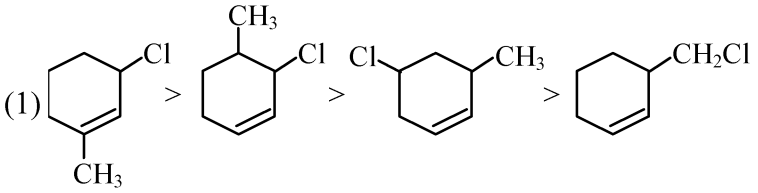
(3) (A) is not correct but (R) is correct

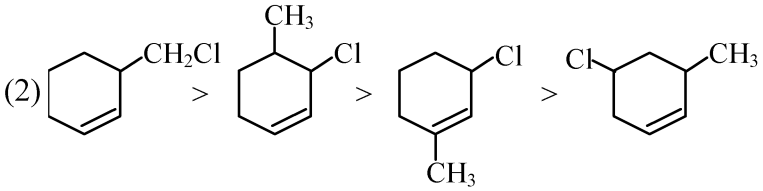
(4) Both (A) and (R) are correct but (R) is NOT the correct explanation of (A)

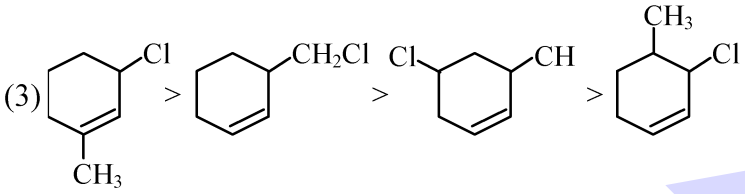
HD0249

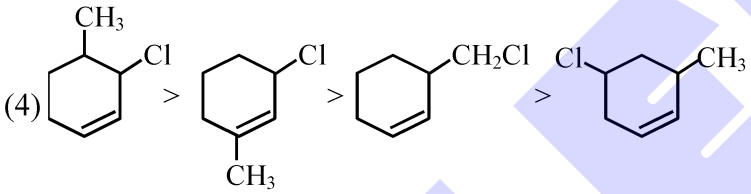
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hallo-Al-& Er.-Hydr.-Carb.-Ox.-Red.-Alld & Ket-(OC)\Eng\01_Halogen Derivatives\04_ExtMains.p65

58. The correct order of reactivity of the given chlorides with acetate in acetic acid is :

(1)  **[JEE-MAIN-2021]**

(2) 

(3) 

(4) 

HD0250

59. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

[JEE-MAIN-2021]

Assertion (A) : Treatment of bromine water with propene yields 1-bromopropan-2-ol.

Reason (R) : Attack of water on bromonium ion follows Markovnikov rule and results in 1-bromopropan-2-ol.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(2) (A) is false but (R) is true.

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) (A) is true but (R) is false

HD0251

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc & Et-Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\01_Halogen Derivatives\04_Exe#Mains.p65

EXERCISE # J-ADVANCE_(OBJECTIVE)

1. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives : **[IIT 1990]**

(A) o-cresol (B) p-cresol

(C) 2,4-dihydroxytoluene (D) Benzoic acid **HD0198**

2. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to **[IIT 1990]**

(A) The formation of less stable carbonium ion (B) Resonance stabilization

(C) The inductive effect (D) sp^2 hybridised carbon attached to the halogen

HD0199

3. 1-Chlorobutane on reaction with alcoholic potash gives : **[ITT 1991]**

(A) 1-butene (B) 1-butanol (C) 2-butene (D) 2-butanol **HD0200**

4. The products of reaction of alcoholic $AgNO_2$ with ethyl bromide are **[IIT 1991]**

(A) Ethane (B) Ethyl nitrite (C) Nitroethane (D) Ethyl alcohol

HD0201

5. Arrange the following compounds in order of increasing dipole moment **[IIT 1996]**

Toluene	m-dichlorobenzene	o-dichlorobenzene	p-dichlorobenzene
I	II	III	IV

(A) I < IV < II < III (B) IV < I < II < III (C) IV < I < III < II (D) IV < II < I < III

HD0202

6. $(CH_3)_3CMgCl$ reaction with D_2O produces:

(A) $(CH_3)_3CD$ (B) $(CH_3)_3OD$ (C) $(CD_3)_3CD$ (D) $(CD_3)_3OD$ **[IIT 1997]**

HD0203

7. Benzyl chloride ($C_6H_5CH_2Cl$) can be prepared from toluene by chlorination with:

(A) SO_2Cl_2 (B) $SOCl_2$ (C) Cl_2,(hν) (D) NaOCl **[IIT 1998]**

HD0204

8. The order of reactivity of the following alkyl halides for a S_N2 reaction is: **[IIT 2000]**

(A) RF > RC > R–Br > R–I (B) R–F > R–Br > R–Cl > R–I

(C) R–Cl > R–Br > RF > RI (D) R–I > RBr > R–Cl > R–F **HD0205**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\Mod #Halo-Alc & Et.-Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\01_Halogen Derivatives\05(i)_Ex#Adv.(Obj).p65"

9. Which of the following has the highest nucleophilicity? **[IIT 2000]**

(A) F^- (B) OH^- (C) CH_3^- (D) NH_2^- **HD0206**

10. An S_N2 reaction at an asymmetric carbon of a compound always gives. **[IIT 2001]**

(A) an enantiomer of the substance (B) a product with opposite optical rotation

(C) a mixture of diasteremoers (D) a single stereoisomer **HD0207**

11. The compound that will react most readily with NaOH to form methanol is **[IIT 2001]**

(A) $(CH_3)_4N^+I^-$ (B) CH_3OCH_3

(C) $(CH_3)_3S^+I^-$ (D) $(CH_3)_3CCl$ **HD0208**

12. Identify the set of reagents / reaction conditions 'X' and 'Y' in the following set of transformation: **[IIT 2002]**

$$CH_3-CH_2-CH_2Br \xrightarrow{X} \text{Product} \xrightarrow{Y} CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$$

(A) X = dilute aqueous NaOH, 20°C; Y = HBr / acetic acid, 20°C

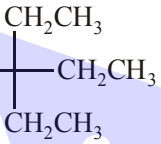
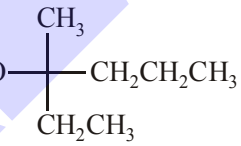
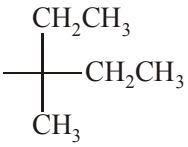
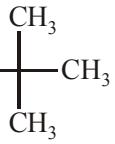
(B) X = concentrated alcoholic NaOH, 80°C; Y = HBr/ acetic acid 20°C

(C) X = dilute aqueous NaOH, 20°C; Y = Br_2 / $CHCl_3$, 0°C

(D) X = concentrated alcoholic NaOH, 80°C; Y = $Br_2/CHCl_3$, 0°C **HD0209**

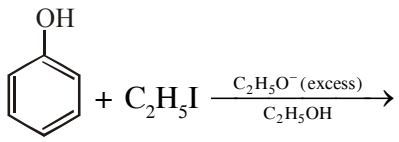
13. CH_3MgBr + Ethyl ester $\rightarrow$ which can be formed as product. **[IIT 2003]**

(excess)

(A)  (B)  (C)  (D) 

HD0210

14. The product of following reaction is **[IIT 2003]**

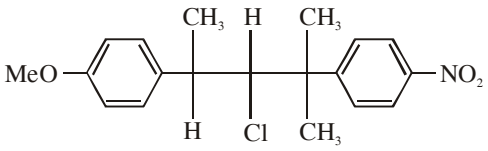


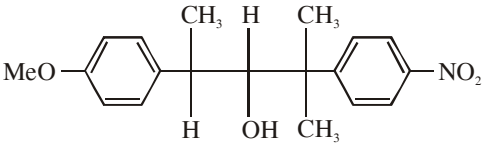
(A) $C_6H_5OC_2H_5$ (B) $C_2H_5OC_2H_5$ (C) $C_6H_5OC_6H_5$ (D) C_6H_5I

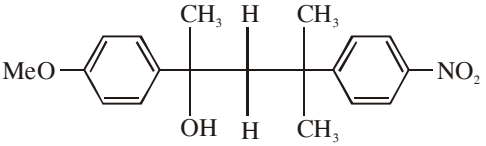
HD0211

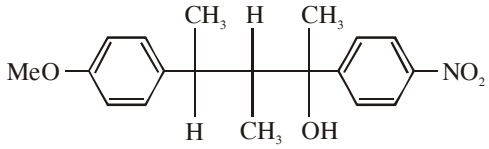
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alk.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\01_Halogen Derivatives\05(i)_Ex#Adv.(Obj.).p65"

15. The following compound on hydrolysis in aqueous acetone will give: **[IIT 2005]**



(K) 

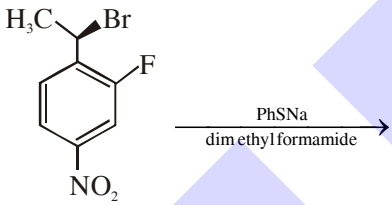
(L) 

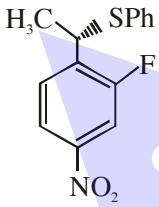
(M) 

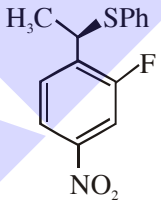
It mainly gives

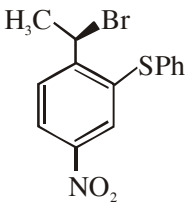
(A) K and L (B) Only K (C)L and M (D) Only M **HD0212**

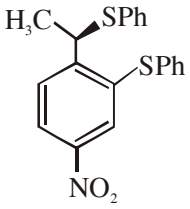
16 The major product of the following reaction is **[IIT 2008]**



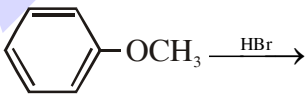
(A) 

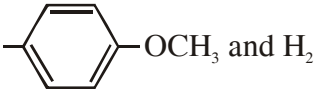
(B) 

(C) 

(D) 

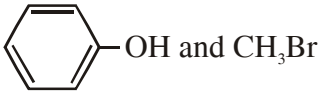
HD0214

17 In the reaction  the products are **[IIT 2010]**

(A) 

(B) 

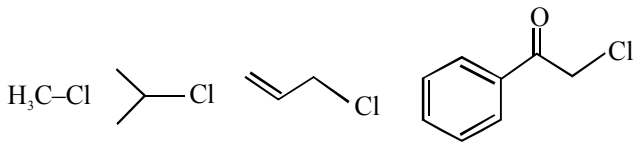
(C) 

(D) 

HD0215

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18. KI in acetone, undergoes S_N2 reaction with each of P, Q, R and S. The rates of the reaction vary as - **[IIT 2013]**

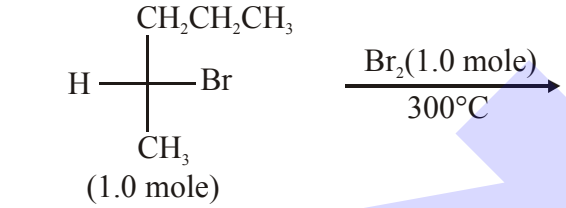


P **Q** **R** **S**

(A) P > Q > R > S (B) S > P > R > Q (C) P > R > Q > S (D) R > P > S > Q

HD0216

19. In the following monobromination reaction, the number of possible chiral products is **[IIT 2016]**

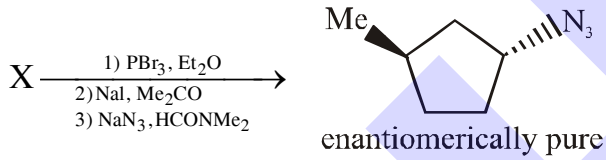


(1.0 mole)

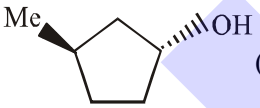
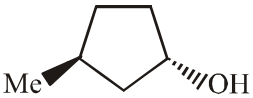
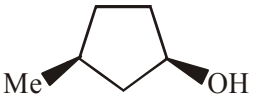
(enantiomerically pure)

HD0218

20. In the following reaction sequence, the correct structure(s) of X is (are) **[IIT-2018]**



enantiomerically pure

(A)  (B)  (C)  (D) 

HD0219

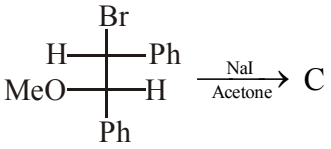
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.& Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\01_Halogen Derivatives\05(i)_Ex#Adv.(Obj.).p65"

EXERCISE # J-ADVANCE_(SUBJECTIVE)

1. An alkyl halide X of formula $C_6H_{13}Cl$ on treatment with potassium tertiary butoxide gives two isomeric alkenes Y and Z (C_6H_{12}). Both alkenes on hydrogenation give 2,3–dimethylbutane. Predict the structures of X, Y and Z. **[IIT 1996]**

HD0220

2. Predict the structure of the intermediates/products in the following reaction sequence – **[IIT 1996]**



HD0221

3. Which of the following is the correct method for synthesising methyl-t-butyl ether and why?

$(CH_3)_3\ CBr + NaOMe \longrightarrow$ or $CH_3Br + NaO\text{-}t\text{-}Bu \longrightarrow$ **[IIT 1997]**

HD0222

4. Write the structures of the products: **[IIT 1998]**

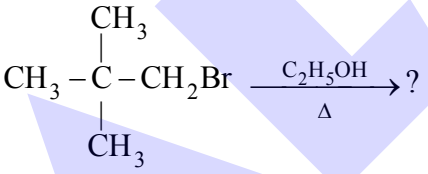
$C_6H_5CH_2CHClC_6H_5 \xrightarrow{Alc.KOH}$ **HD0223**

5. (a) $C_6H_5CH_2\underset{\displaystyle C_6H_5}{\underset{|}{C}}HCl \xrightarrow[Heat]{alc.\ KOH} A + B$ Write structures of (A) and (B).

(b) $(CH_3)_2CHOCH_3 \xrightarrow[Heat]{HI(excess)} A + B$ Write structures of A and B. **[IIT 1998]**

HD0224

6. What would be major product? **[IIT 2000]**

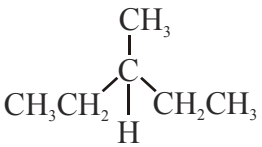


HD0225

7. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is **[IIT 2011]**

HD0226

8. The maximum number of isomers (including stereoisomers) that are possible on mono-chlorination of the following compounds, is **[IIT 2011]**



HD0227

9. The maximum number of possible isomers (including stereoisomers) which may be formed on mono-bromination of 1-methylcyclohex-1-ene using Br_2 and UV light is ____ **[IIT 2021]**

HD0252

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ANSWER-KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	D	D	C	C	B	D	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	A	D	B	A	D	C	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	D	A	B	C	B	C	D	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	B	C	A	A	C	D	A	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	C	A	C	A	C	D	D	B	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	C	A	C	A	D	C	B	B	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	D	A	B	A	C	D	B	B	C
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	B	C	C	A	B	A	A	D	C	B
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	C	C	C	B	D	C	B	A	C	C
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	A	C	B	B	A	B	D	B	B	D

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B,C	A,B	B,D	B,D	A,C	A,C,D	A,C	A,B,D	A,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,D	A,C,D	A,C,D	A,B,C	A,C,D	B,D	B,D	A,B	A,B,C	C,D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B,C	B,C,D	A,B,D	A,B	A,C	A,B,D	A,B,C,D	A,B,C,D	B	A,B,C
Que.	31									
Ans.	A,C									

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EXERCISE # S-1

1. (C) **2.** (C) **3.** (B)

4. (A) → P, R ; (B) → P, Q ; (C) → P ; (D) → P, Q

5. (A)→Q ; (B)→P ; (C)→S ; (D)→R

6. (A) → S ; (B) → Q ; (C) → R ; (D) → P

7. (A) → S ; (B) → R, S ; (C) → R ; (D) → P, Q

8. (A) → P, Q, R, S ; (B) → P, R ; (C) → P, T ; (D) → Q, S

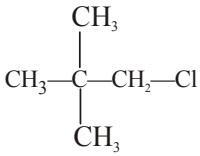
9. (A) → Q, S ; (B) → P, R, S ; (C) → P, R, S ; (D) → P, Q, R, S

10. (A) → S, T ; (B) → P , S, T ; (C) → U ; (D) → Q ; (E) → T, U

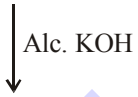
11. (A) → R ; (B) → Q, S ; (C) → P, Q ; (D) → Q, S

12. (A) → P ; (B) → P ; (C) → Q ; (D) → R

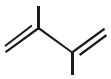
13. C–I bond being less stable than C–Cl bond and thus on heating heterolytic cleavage of C–I form I^- which gives yellow precipitate with $AgNO_3$.

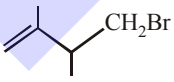
14. **X is** 

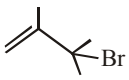
15. **Molecule A, $C_6H_{11}Br$ has 1 unsaturation**



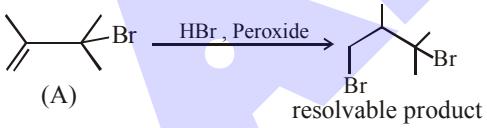
A single possible product , it suggests a symmetrical arrangement

So, E = 

There are only two possibilities of A 

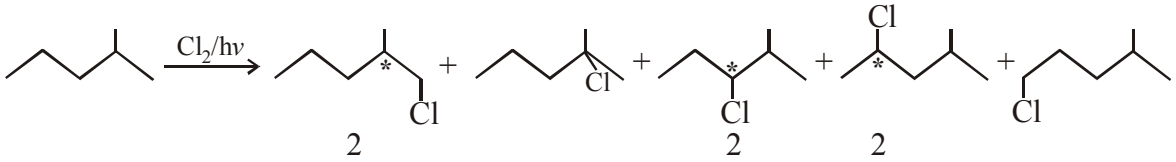
(I) or 

(II) structure I can be resolved while structure II cannot be resolved so 'A' :



(A) resolvable product (C)

16. **Both step is exothermic with HBr**

17. 

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18. **(a)** **Ph—CH = $CHCH_3$ + HBr** $\longrightarrow$ Ph—CH(Br)—CH_2—CH_3

(b) H_3C, H_3C, C=C, CH_3, H + HI $\longrightarrow$ CH_3—C(CH_3)(I)—CH_2—CH_3

(c) CH_3 + HBr $\xrightarrow{\text{Peroxide}}$ CH_3, Br

(d) CH_3, CH_2–CH_3 + HCl $\longrightarrow$ CH_3, Cl, CH_2–CH_3

19. (A).7gm $\xrightarrow{Br_2,\ 2\ gm}$ H—C(CH_3)(Br)—C(H)(Br)—CH_3

(A).7gm $\xrightarrow{HBr}$ (B) Br

(A).7gm $\xrightarrow{HBr}$ (B) Br

20. **(a)** H, CH_2, CH_3 $\xrightarrow{HCl}$ H, Cl, CH_3, CH_3 + H, CH_3, Cl, CH_3

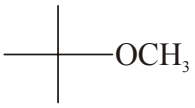
(b) CH_3 $\xrightarrow[CCl_4]{Br_2}$ Br, Br, CH_3 + Br, Br, CH_3

(c) CH_3, H—C(Br)—C(H)(Br)—Et + Enantiomer

21. The elimination of HI (or DI) in presence of strong base shows E2 elimination. The rate determining step involves breaking up of C – H (or C–D) bond. The C–D bond being stronger than C–H and thus elimination is faster in case of $CH_3 - CH_2I$.

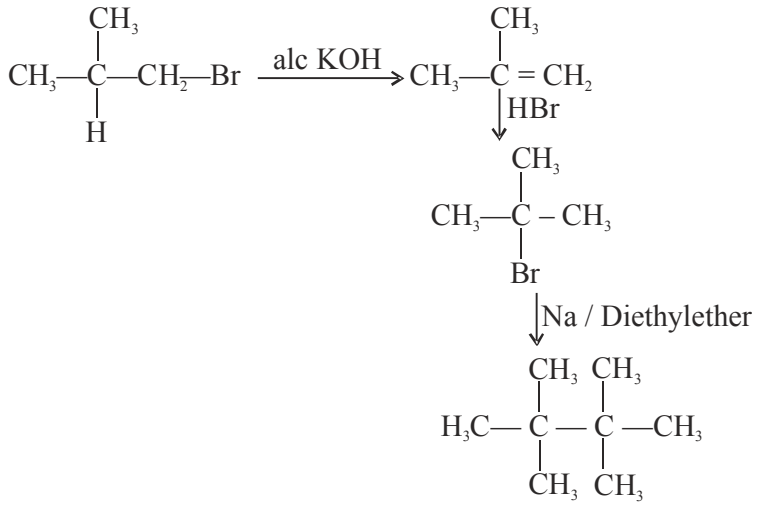
22. In second compound π bonds are conjugated so due to resonance given product is formed as major product.

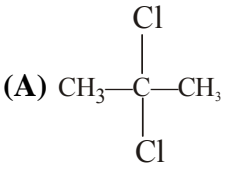
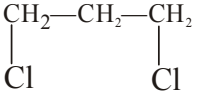
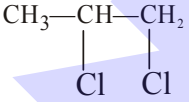
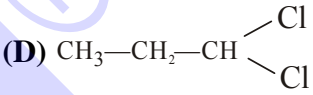
23. O $\xrightarrow[\Delta]{HI(excess)}$ I, I

24. **(a)**  **(b)** 

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25.



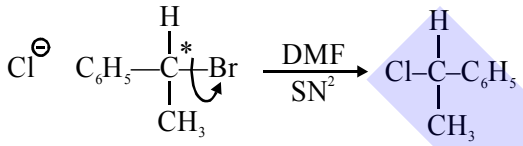
26. **(A)**  **(B)**  **(C)**  **(D)** 

EXERCISE # J-MAINS

1. **(3)** **2.** **(1)** **3.** **(4)** **4.** **(2)**

5. **(2)** **6.** **(4)**

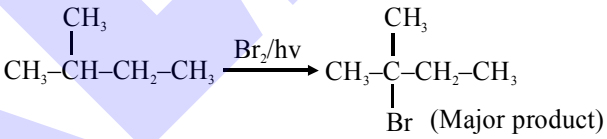
7. **(2)**

Sol. 

inverted product

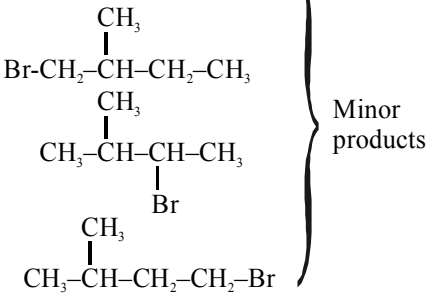
8. **(1)**

Sol.



(Major product)

2-Bromo-2-methyl butane



Minor products

relectivity ratio for bromination is

1° : 2° : 3° : : 1 : 82 : 1600

Hence 3° product will be major product.

9. **(2)** **10.** **(2)**

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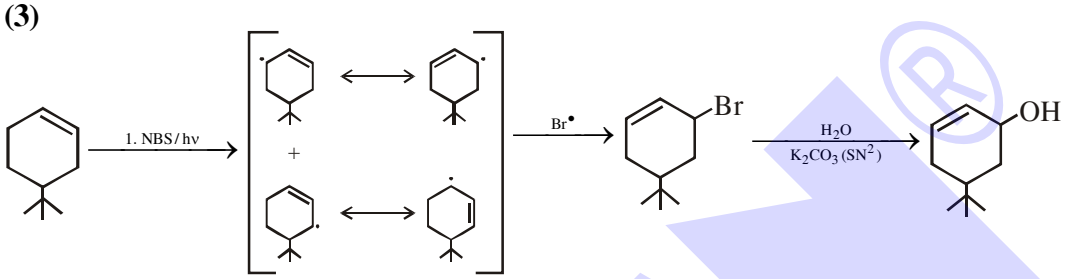
11. (2)

Sol. $C_2H_5CH_2\underset{\underset{Cl}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3 \xrightarrow[CH_3OH]{NaOCH_3}$

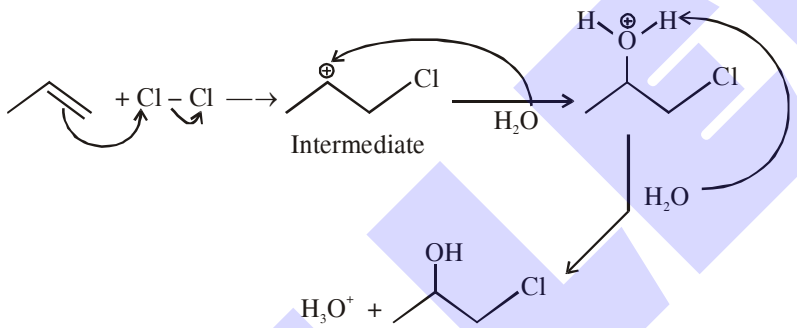
possible mechanism which takes place is E^2 & SN^1 mechanism. Hence possible products are.

$C_2H_5CH_2\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OCH_3$ (SN^1) $\quad C_2H_5CH_2C(CH_3)=CH_2$ $\quad C_2H_5CH=C(CH_3)-CH_3$ (E^2)

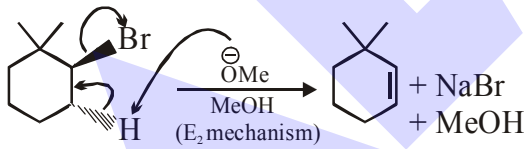
12. (3)

Sol. 

13. (3)

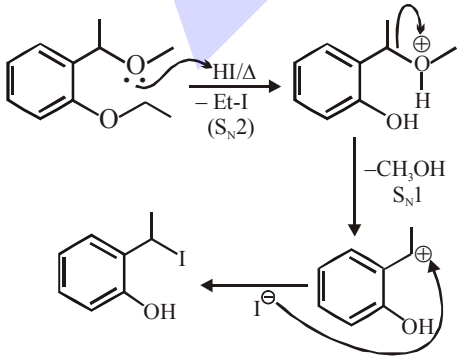
Sol. 

14. (2) **15. (1)** **16. (4)** **17. (2)** **18. (1)**

Sol. 

Reaction is dehydrohalogenation E^2-elimination reaction. Elimination takes place in single step and proceed by formation of transition state from anti position.

19. (3)

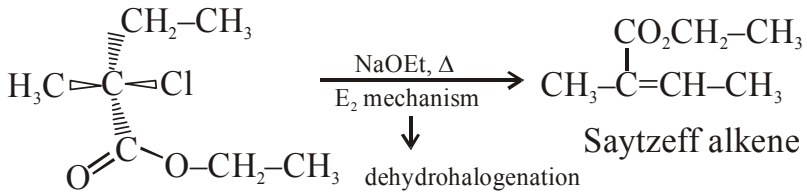
Sol. 

It is nucleophilic substitution reaction.

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20. (3)

Sol.

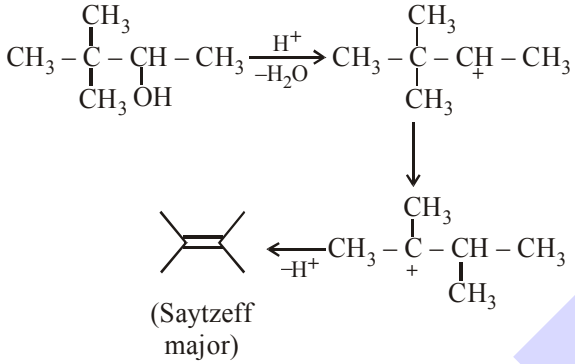


21.	**(4)**	**22.**	**(4)**	**23.**	**(1)**	**24.**	**(4)**
25.	**(4)**	**26.**	**(3)**	**27.**	**(3)**	**28.**	**(3)**
29.	**(1)**	**30.**	**(3)**	**31.**	**(2)**	**32.**	**(2)**

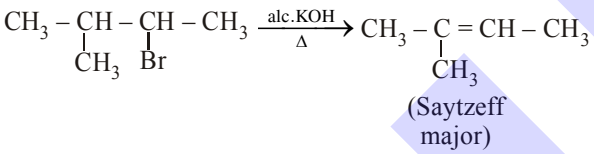
33. (1)

Sol.

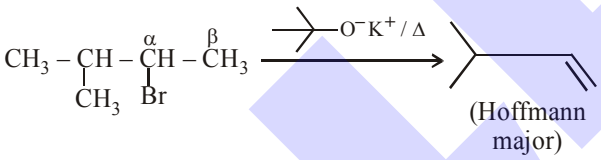
(a)



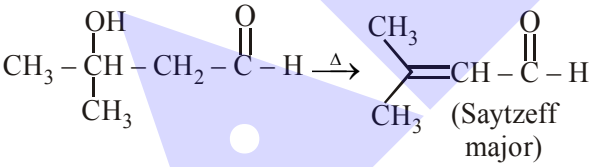
(b)



(c)



(d)



$(CH_3)_3O^-K^+$ is incorrect representation of potassium tert-butoxide [$(CH_3)_3CO^-K^+$].

So it is possible that it can be given as Bonus

34. (3)

Sol. Bond length order in carbon halogen bonds are in the order of C – F < C – Cl < C – Br < C – I

Hence, Bond energy order

C – F > C – Cl > C – Br > C – I

35. (3)

Sol. Reactivity D > B > C > A

Carbocation formed from D is most stable

Carbocation formed from A is least stable

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36.	**(1)**	**37.**	**(3)**	**38.**	**(4)**	**39.**	**(3)**
40.	**(2)**	**41.**	**(3)**	**42.**	**(1)**	**43.**	**(5)**
44.	**(4)**	**45.**	**(1)**	**46.**	**(4)**	**47.**	**(3)**
48.	**(4)**	**49.**	**(2)**	**50.**	**(2)**	**51.**	**(4)**
52.	**(1)**	**53.**	**(2)**	**54.**	**(4)**	**55.**	**(2)**
56.	**(4)**	**57.**	**(2)**	**58.**	**(1)**	**59.**	**(3)**

EXERCISE # J-ADVANCE_(OBJECTIVE)

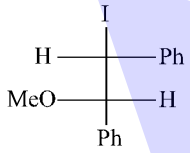
1.	**(D)**	**2.**	**(B,D)**	**3.**	**(A)**	**4.**	**(C)**	**5.**	**(B)**
6.	**(A)**	**7.**	**(A,C)**	**8.**	**(D)**	**9.**	**(C)**	**10.**	**(D)**
11.	**(A)**	**12.**	**(B)**	**13.**	**(D)**	**14.**	**(B)**	**15.**	**(A)**
16.	**(A)**	**17.**	**(D)**	**18.**	**(B)**	**19.**	**(5)**		
20.	**(B)**								

Sol. **X** $\xrightarrow[(3)NaN_3, HCONMe_2]{(1)PBr_3Et_2O \quad (2)NaI, Me_2C=O}$ 

all the three reaction are S_{N^2} **so X is** 

EXERCISE # J-ADVANCE_(SUBJECTIVE)

1. **(X) :** $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{CH_3}{|}}{C}}-\overset{\overset{CH_3}{|}}{CH}-CH_3$ **; (Y) :** $CH_2=\overset{\overset{CH_3}{|}}{C}-\overset{\overset{CH_3}{|}}{CH}-CH_3$ **; (Z) :** $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-\overset{\overset{CH_3}{|}}{CH}-CH_3$

2.  **(NGP due to MeO, so retention of configuration)**

3. $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-Br + CH_3O^- \xrightarrow{E2} CH_2=C\begin{matrix}CH_3\\CH_3\end{matrix}$ **;** $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-O^- + CH_3Br \xrightarrow{S_{N^2}} CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OCH_3$

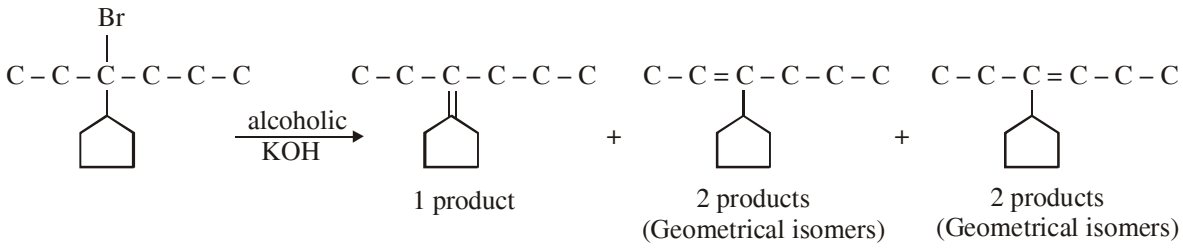
4. $C_6H_5-CH_2-\underset{\underset{Cl}{|}}{CH}-C_6H_5 \xrightarrow[E2]{Alc.KOH}$ **$C_6H_5-CH=CH-C_6H_5$**

5. **(a) Cis and trans forms of stibene $C_6H_5CH=CHC_6H_5$; (b)** $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>CHI + CH_3I$

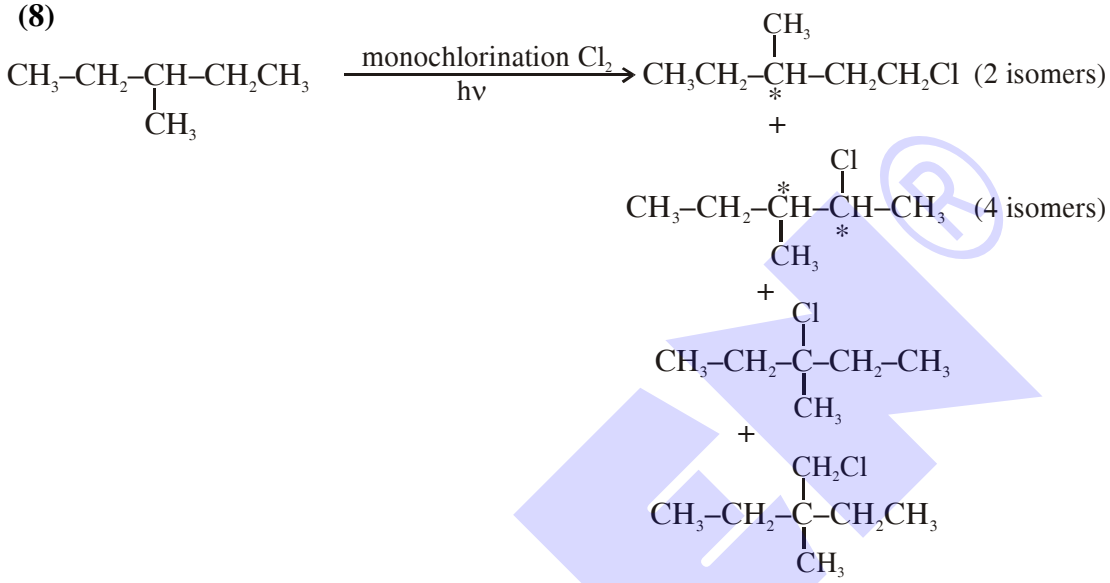
6. $CH_3-\overset{\overset{CH_3}{|}}{C}=CHCH_3$

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7. **(5)**



8. **(8)**



9. **Both answer 9 and 13 are correct according to IIT**

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